Low-Overhead Surface Code with Delayed Atom-Loss Detection via Pauli Envelope

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Atom loss is a major error source in neutral-atom quantum computers, accounting for over 40% of the total physical errors in recent experiments. We consider the loss model of current alkali-atom architectures, in which a loss is revealed only by the destructive readout at the end of the atom's life cycle and removes the gates acting on the lost atom, as opposed to erasures heralded mid-circuit. The nonlinear and correlated nature of this error poses significant challenges: current syndrome extraction circuits require additional overhead or sacrifice loss tolerance, and existing decoders are computationally inefficient, suboptimal, or lack provable guarantees. To address these challenges, we propose the *Pauli Envelope* framework, which bounds the effect of atom loss with low-weight, efficiently computable Pauli approximations, generalizing existing loss-to-Pauli methods and enabling rigorous analysis. Guided by this framework, we design an improved atom-replenishing syndrome extraction circuit, the Mid-SWAP syndrome extraction, which achieves the optimal loss distance for rotated surface codes at the space-time cost of standard syndrome extraction, whereas previous circuits reach it only with additional overhead or only in the absence of Pauli errors. We also propose two decoders: an *Envelope-MLE decoder* achieving the optimal loss distance $d_{\text{loss}} \sim d$, and an *Envelope-Matching decoder* achieving $d_{\text{loss}} \sim 2d/3$ via Minimum-Weight Perfect Matching (MWPM), surpassing the previous best ($d_{\text{loss}} \sim d/2$) and readily integrating with fast correlated decoding techniques for transversal logical circuits. Circuit-level simulations demonstrate over 40% higher thresholds and 30% higher effective distances compared with existing methods in the loss-dominated regime. Moreover, we explore correlated atom loss and show that it is easier to correct than independent loss, with thresholds rising from 5.15% to 7.82%. Remarkably, our Envelope-MLE decoder improves the error suppression factor of a hybrid MLE–machine-learning decoder from $\Lambda = 2.14$ to $\Lambda = 2.24$ on recent experimental data.

I. INTRODUCTION

Quantum computing promises transformative advances in cryptography [1], materials science [2], and optimization [3], yet realizing this potential requires fault-tolerant operation through quantum error correction [4]. The most widely studied quantum error-correcting code for achieving fault tolerance is the surface code [5, 6]. It is known to possess a high physical error threshold under various noise models [7, 8] and can be implemented with nearest-neighbor connectivity. Among the leading platforms for fault-tolerant quantum computing, neutral-atom quantum computers have demonstrated exceptional scalability and gate fidelity [9–13]. However, atom loss remains one of the key challenges for achieving fault tolerance in these systems, accounting for more than 40% of the total physical errors in recent experiments [10].

In this paper we consider the loss model of current alkali-atom architectures [9, 10]: a loss is revealed only by the destructive, loss-resolving readout at the end of the atom's life cycle. Between the loss and this readout, every gate scheduled on the lost atom acts as the identity; in particular, a two-qubit gate leaves its partner qubit untouched, as if the gate had been removed from the circuit. The missing entangling gates cause correlated Clifford errors that propagate through the computation [14–17]. This delayed detection is specific to the platform and is not the only possible loss model: platforms with heralded mid-circuit leakage detection, demonstrated with alkaline-earth atoms [18–20], lead instead to the erasure model [21], which we use only as a point of comparison (Section II). Compared with erasures, gate-removing loss with delayed detection exhibits two features that make it particularly challenging. First, atom loss persists until replenishment, causing errors to accumulate across multiple gates and even multiple syndrome extraction rounds. Specialized syndrome extraction circuits are therefore needed to localize loss effects in both time and space. Second, unlike Pauli errors, which propagate and combine linearly, atom loss behaves nonlinearly. Multiple atom losses may produce detector patterns that cannot be derived by composing individual loss patterns, posing significant challenges for decoder design and analysis. To address these challenges, we develop the *Pauli Envelope* framework to linearize the nonlinear loss effects into tractable Pauli errors. This framework rigorously characterizes how atom loss effects can be bounded by low-weight Pauli errors, guiding the design of syndrome extraction circuits and decoding algorithms with provable performance guarantees. Unlike prior heuristic approximations, the Pauli envelope provides two key guarantees: (1) *correctness*: if the Pauli envelope is decoded correctly, the original atom-loss events are guaranteed to be decoded correctly; and (2) *low weight*: the

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Pauli envelope has low weight for loss errors, leading to stronger distance guarantees.

Guided by this framework, we make the following contributions:

(1) Syndrome extraction circuit: We introduce the *Mid-SWAP syndrome extraction*, which reduces the propagation of loss effects with no additional space-time overhead compared to the conventional rotated surface code syndrome extraction circuit. To quantify this improvement, we use the *loss distance* d_{loss}, which characterizes how many simultaneous atom losses a code-decoder pair can tolerate before a logical error occurs (see Definition 1 for the formal definition). This circuit achieves $d_{\text{loss}} \sim d$, where d is the code distance (we write $d_{\text{loss}} \sim f(d)$ to denote $d_{\text{loss}} = f(d) \pm O(1)$), which is optimal for a distance-d surface code and significantly outperforms the SWAP syndrome extraction [14], which we will show only achieves $d_{\text{loss}} \sim d/2$.

(2) Optimal-distance decoder: We propose the *Envelope-Most-Likely-Error decoder (Envelope-MLE decoder)*, which achieves $d_{\text{loss}} \sim d$ for the Mid-SWAP syndrome extraction. This significantly outperforms the *Average-MLE decoder* of [14], which we will show only achieves $d_{\text{loss}} \sim d/2$. The key insight is that each atom loss triggers exactly one detector pattern in our Pauli envelope formulation, imposing an *exclusivity* constraint that the Average-MLE decoder fails to enforce.

(3) Efficient decoder: Inspired by the exclusivity constraint, we design the *Envelope-Matching decoder* for efficient decoding. Instead of changing edge weights to a constant, which becomes negligible compared with Pauli error weights when $p_{\text{pauli}} \to 0$, as done in the *Marginal-Matching decoder* by Gu et al. [22, 23], we rescale the affected edge weights by a constant factor to discourage selecting multiple edges in the same envelope, approximately enforcing the exclusivity constraint. This decoder achieves $d_{\text{loss}} \sim 2d/3$ for the Mid-SWAP syndrome extraction, outperforming the $d_{\text{loss}} \sim d/2$ of the Marginal-Matching decoder. As an MWPM-based decoder, the Envelope-Matching decoder naturally integrates with fast correlated decoding techniques for transversal logical circuits [24–26].

Table I summarizes the theoretical performance of our proposed decoders (Envelope-MLE decoder and Envelope-Matching decoder) compared to existing approaches across several atom-loss correction settings, including our newly proposed Mid-SWAP syndrome extraction. Unless explicitly stated otherwise, we consider rotated surface codes throughout.

Before this work, distance-d scaling under a similar loss model had been reported for the toric code [15] and for a measurement-based architecture [16]; Appendix F derives the former as a special case of our framework. Both, however, incur a higher space-time cost than the rotated surface code on neutral-atom platforms; for instance, the unrotated toric code uses twice as many data qubits at the same distance [27].

On rotated surface codes, numerical studies have shown that the loss distance can approach $\sim d$ in the absence of Pauli errors, using SWAP replenishment [14] or additional loss-detection units [28], and Ref. [14] proposes a heuristic decoder for the general case. However, pure-loss scaling does not by itself establish the same tolerance jointly with Pauli errors (Section II), and the heuristic decoder comes without a guarantee: as we show in Appendix E 2, it is limited to $d_{\text{loss}} \sim d/2$ once Pauli errors are present.

Loss Dist. \ **Circuit** / **Decoder**	SWAP	**Mid-SWAP**	Teleportation	Unrotated SWAP [15]
Average-MLE	$d/2$	$d/2$	$d/2$	d
Marginal-Matching	$d/2$	$d/2$	$d/2$	d
Envelope-MLE	$d/2$	**d**	d	d
Envelope-Matching	$d/2$	**2d/3**	$2d/3$	d
Space-Time Cost	1×	1×	$\geq 2\times$	2×

TABLE I: Theoretical performance comparison of decoders and syndrome extraction circuits with atom-loss replenishment. The bold entries are introduced in this paper. All distances are up to additive constants. Space-time cost is the number of qubits multiplied by the duration of one syndrome extraction round, relative to the standard rotated surface code circuit. Teleportation-based syndrome extraction is briefly discussed in Appendix H. Unrotated surface codes will be discussed as a special case of HGP codes in Appendix F.

In contrast, we give a formal definition of loss distance that admits any combination of loss and Pauli errors (Definition 1). Using the Pauli envelope, we prove that the Mid-SWAP syndrome extraction with the Envelope-MLE decoder attains $d_{\text{loss}} \sim d$ at the space-time cost of the standard circuit; this is the largest loss distance possible, since d losses along a logical string erase the encoded information. The efficient Envelope-Matching decoder attains $\sim 2d/3$ with the same circuit.

We evaluate the performance of our proposed decoders and syndrome extraction circuit using a circuit-level noise model. In the loss-dominated regime, our approach achieves over 40% higher thresholds and 30% higher effective distances than prior approaches, and we show that correlated atom loss is *easier* to correct than independent loss. Fig. 1 previews the size of the effect at pure loss: changing only the circuit or only the decoder lowers the logical error rate by a factor between 3.0 and 8.4, and the combination by 13.7× at $d = 9$ and 26.3× at $d = 11$.

This paper is organized as follows: Section II reviews the atom-loss error model and essential background for quantum error correction. Section III introduces the Pauli envelope framework. Section IV presents the Mid-SWAP syndrome extraction circuit. Section V describes the Envelope-MLE decoder. Section VI presents the Envelope-Matching decoder. Section VII provides nu-

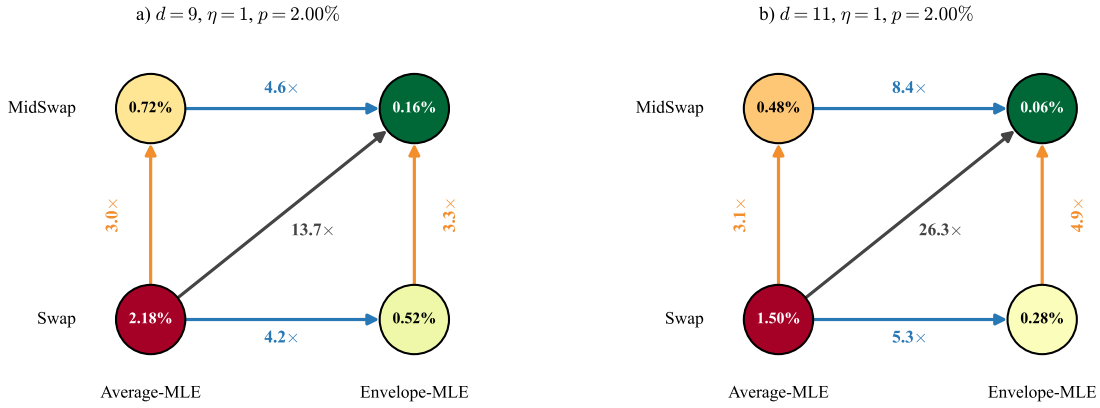


FIG. 1: Logical error rate of the four decoder–circuit combinations at pure loss ($\eta = 1$) and $p = 2.0\%$ for (a) $d = 9$ and (b) $d = 11$, all with identical noise model and sampling. Edge labels give the error-reduction factor for changing only the decoder (horizontal), only the circuit (vertical), or both (diagonal).

merical evaluation.

II. ATOM LOSS

Various terms are used in the literature to describe errors where qubits leave the computational subspace [29–31]. We distinguish them by two criteria, summarized in Table II: when the error is detected and how it affects subsequent gates. We use *atom loss* for an error that is detected only at measurement and removes the subsequent gates on the lost atom, and *erasure* for an error that is detected immediately and turns the subsequent gates into maximally depolarizing errors.

TABLE II: Comparison of erasure errors and atom loss.

	Erasure Error	Atom Loss
Detection timing	Instantaneous	Upon measurement
Effects	Maximum depolarizing	Gate-removing
Distance	d	$\sim \mathbf{d}$

Hardware model. We model a neutral-atom computer of the alkali-atom architecture family [9, 10] as a collection of atoms with finite life cycles: an atom is created in $|0\rangle$, acted on by single- and two-qubit gates, and finally measured. Following the state-selective readout demonstrated on these platforms [9, 10], a measurement has three outcomes: $|0\rangle$, $|1\rangle$, or lost. The readout is destructive and ends the atom's life cycle, so it cannot check for loss mid-circuit; when it reports a loss, we only learn that the atom was lost at some point between its creation and its measurement. Measured atoms are discarded, and lost atoms are replaced only by creating fresh atoms in $|0\rangle$. This is the least demanding setting capable of fault-tolerant quantum computation [9], and we assume it throughout this paper. Fault tolerance then requires syndrome extraction circuits that regularly measure and replenish every atom, including the data qubits.

Loss model. Following [14], we model atom loss as a gate-removing error: after an atom is lost, every subsequent gate acting on it, single- or two-qubit, is removed from the circuit, i.e., acts as the identity (Fig. 2). Gates that do not involve the lost atom are unaffected, and the loss itself does not spread to other atoms. A lost atom's partners only miss their interactions with it until the lost atom is measured and replenished.

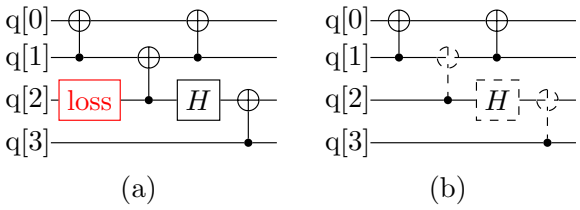


FIG. 2: Atom loss as a gate-removing error. (a) An atom loss occurs at the beginning of the circuit on qubit $q[2]$. (b) The atom loss removes all subsequent gates acting on qubit $q[2]$ (dashed); gates that do not involve $q[2]$, such as the second $CNOT$ between $q[0]$ and $q[1]$, are unaffected, because the loss itself does not spread to other atoms.

In contrast to the alkali-atom architecture considered here, alkaline-earth atoms allow leakage out of the qubit subspace to be detected mid-circuit without disturbing the remaining qubits [18–20, 32, 33]; such heralded leakage is an erasure error in the sense of Table II and is outside the scope of this paper.

Detectors and nonlinearity. Decoding relies on the concept of *detectors*. Detectors are linear combinations of measurements that are deterministic under ideal execution of a Clifford circuit. For example, in a surface

code quantum memory, a detector is typically the XOR of two consecutive syndrome measurements at the same stabilizer location: under ideal execution, both measurements yield the same result, so their XOR is always 0. During noisy execution, we obtain a detector pattern $\mathcal{D}$, indicating the set of detectors that are different from the ideal outcome, and a decoder finds an error configuration that can explain this pattern.

A *detector error model* (DEM) is a collection of elementary error mechanisms, each with an associated probability of occurrence and the detector pattern it triggers. For Pauli noise in Clifford circuits, errors compose linearly: the combined detector pattern of multiple Pauli errors is the sum (modulo 2) of their individual detector patterns, which enables a compact DEM description.

Atom loss breaks this additivity because it introduces Clifford errors. The interactions between Clifford errors are much more complicated than those of Pauli errors. As a result, the detector pattern resulting from multiple loss events generally cannot be obtained by summing the detector patterns of individual loss events in isolation.

Fig. 3 shows a minimal example: without error, or losing either the first or the second atom alone, the final qubit yields $|1\rangle$ in panels (a)–(c), while with both losses it yields $|0\rangle$ in panel (d).

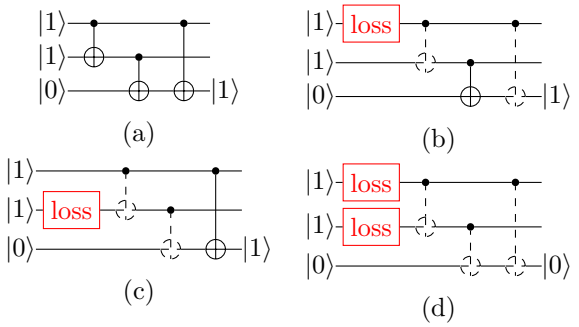


FIG. 3: The composition of atom loss is nonlinear.

This nonlinearity poses a significant challenge for decoding: even if we know exactly which atoms were lost and when, we do not have a simple closed-form expression for the distribution of measurement outcomes, as we do for Pauli errors. This makes rigorous analysis of atom-loss decoding performance challenging.

Loss distance. To systematically analyze and compare the performance of different syndrome extraction circuits and decoders, we define the *loss distance* through the following threshold-style scaling:

Definition 1 (Loss Distance)**.** *A distance-d code and decoder pair has loss distance d_{loss} if there exists a constant $p_{\text{th}} > 0$, independent of d, such that the logical error rate P_{LER} satisfies*

$$P_{\text{LER}} = O\left(p_{\text{th}}^{-d}(p_{\text{loss}}^{d_{\text{loss}}} + p_{\text{pauli}}^{d/2})\right),$$

where p_{loss} and p_{pauli} denote the physical loss and Pauli error rates, respectively, and the bound holds uniformly for sufficiently small values of both rates.

The exponent $d/2$ gives a uniform bound including mixed loss–Pauli terms. In other words, if a cross-term such as $p_{\text{loss}}^{d/2}p_{\text{pauli}}$ appears in the logical error rate, then $d_{\text{loss}} \leq d/2 + O(1)$. In contrast, cross-terms such as $p_{\text{loss}}^{d/2}p_{\text{pauli}}^{d/4}$ are consistent with $d_{\text{loss}} = d$, since $p_{\text{loss}}^{d/2}p_{\text{pauli}}^{d/4} \leq p_{\text{loss}}^{d} + p_{\text{pauli}}^{d/2}$. For simplicity, we write the Pauli exponent as $d/2$ instead of the exact value $(d+1)/2$ of an odd-distance code. This makes the bound slightly looser and does not affect d_{loss}, which we state only up to additive constants.

III. PAULI ENVELOPE

In this section, we formalize the Pauli envelope, prove its key properties, and show how to construct Pauli envelopes for atom loss. A summary of the notation used throughout the paper is given in Appendix N.

A. Formalization of Pauli Envelope

The key idea behind the Pauli envelope is simple: although atom loss behaves nonlinearly, its effect on the decoder can be *bounded* by a set of Pauli errors. If a decoder can correctly handle all errors in this bounding set, it will also correctly decode the original atom loss. This reduces the complex problem of decoding atom loss to the well-understood problem of decoding Pauli errors.

We now formalize this intuition. Consider a Clifford circuit $\mathcal{C}$ with Pauli error configuration p and loss configuration l, where l is a set of space-time locations at which atoms are lost and p assigns a Pauli operator to each faulty space-time location. If an atom is lost more than once in a configuration, we ignore subsequent loss events. We write $p \oplus p'$ for the composition of two Pauli error configurations, i.e., the product of their Pauli operators at each location; global phases never affect detectors or observables and are ignored. For loss configurations, $l \oplus l'$ denotes the union of their loss locations.

Let $(\mathcal{D}(p,l), \mathcal{O}(p,l))$ denote the random detector-observable pair produced by running $\mathcal{C}$ with error p and loss l. We omit the circuit subscript $\mathcal{C}$ when it is clear from context. Measuring an atom m yields a three-outcome physical readout $\tilde{b}_m \in \{0, 1, \perp\}$, where $\perp$ indicates that the atom is lost. Detectors and observables are computed from binary bits b_m: $b_m = \tilde{b}_m$ if the atom is present, and b_m is assigned a uniformly random bit if $\tilde{b}_m = \perp$. This random assignment is a modeling convention for the missing measurement bit and is independent of the Pauli envelope introduced below. Define $\mathcal{S}(p,l)$ as the support of this pair, i.e., all detector-observable pairs with nonzero probability at fixed p, l. For sets P, L of

configurations, define

$$\mathcal{S}(P, L) = \bigcup_{p \in P,\, l \in L} \mathcal{S}(p, l).$$

When an argument is a single configuration, it stands for its singleton set; in particular, $\emptyset$ denotes the no-loss configuration. Probabilities below also average over circuit measurements and assigned bits.

The *loss-resolving readout* r collects the bits $r_m = 1$ if $\tilde{b}_m = \perp$ and $r_m = 0$ otherwise over all measurements. With perfect loss detection, it is a deterministic function of the loss configuration, denoted $r = \mathcal{R}(l)$, and readouts compose by taking the union of the atoms they report as lost, $r \oplus r'$. Multiple loss configurations (differing in *when* those atoms were lost) may produce the same loss-resolving readout, so we define $\mathcal{L}(r) = \{l : \mathcal{R}(l) = r\}$ as the preimage.

Definition 2 (Pauli Envelope)**.** *A loss configuration l has Pauli envelope E, where E is a set of Pauli error configurations, if for all Pauli error configurations p,*

$$\mathcal{S}(p, l) \subseteq \mathcal{S}(p \oplus E, \emptyset),$$

where $\emptyset$ denotes no atom loss and $p \oplus E$ denotes the composition of p with errors from E, i.e., $p \oplus E = \{p \oplus e : e \in E\}$.

Similarly, we say a readout r has Pauli envelope E if $\mathcal{S}(p, \mathcal{L}(r)) \subseteq \mathcal{S}(p \oplus E, \emptyset)$ for all p.

In other words, any detector-observable pair that can arise from a Pauli error p combined with any loss configuration producing readout r can also arise from the Pauli errors $p \oplus E$ with no atom loss.

The Pauli envelope ensures that if we can correctly decode the Pauli envelope, we can decode the atom losses correctly.

Lemma 1 (Sufficiency of Decoding Pauli Envelope)**.** *Let $\mathcal{C}$ be a Clifford circuit suffering from a Pauli error configuration p and a loss configuration with readout r. Suppose that r has Pauli envelope E. Let Dec be a deterministic decoder that maps detector patterns and loss-resolving readouts to logical observable predictions.*

If $\mathsf{Dec}(\cdot, r)$ correctly decodes all detector-observable pairs in $\mathcal{S}(p \oplus E, \emptyset)$, i.e., for all $(\mathcal{D}, \mathcal{O}) \in \mathcal{S}(p \oplus E, \emptyset)$, we have $\mathsf{Dec}(\mathcal{D}, r) = \mathcal{O}$, then $\mathsf{Dec}(\cdot, r)$ also correctly decodes all detector-observable pairs in $\mathcal{S}(p, \mathcal{L}(r))$.

Proof. Since r has Pauli envelope E, by definition we have

$$\mathcal{S}(p, \mathcal{L}(r)) \subseteq \mathcal{S}(p \oplus E, \emptyset).$$

Therefore, if $\mathsf{Dec}(\cdot, r)$ correctly decodes all pairs in $\mathcal{S}(p \oplus E, \emptyset)$, it also correctly decodes all pairs in $\mathcal{S}(p, \mathcal{L}(r))$. □

Next, we relate the logical error rate with atom loss to the failure probability with the Pauli envelope. We first extend the definition of logical error rate to include atom loss.

Definition 3 (Logical Error Rate with Atom Loss)**.** *Let $\mathcal{C}$ be a Clifford circuit with the following error model:*

- *Pauli error configurations are sampled from distribution $\mathcal{P}_p$.*
- *Loss configurations are sampled from distribution $\mathcal{P}_l$.*

The logical error rate $P_{\mathrm{LER}}(\mathcal{C}, \mathcal{P}_l, \mathcal{P}_p, \mathsf{Dec})$ is defined as

$$P_{\mathrm{LER}} = \Pr_{\substack{l \sim \mathcal{P}_l \\ p \sim \mathcal{P}_p}} \left[\mathsf{Dec}(\mathcal{D}(p, l), \mathcal{R}(l)) \neq \mathcal{O}(p, l)\right] \tag{1}$$

We further define the failure probability with Pauli envelope.

Definition 4 (Failure Probability with Pauli Envelope)**.** *Let $\mathcal{C}$ be a Clifford circuit with the following error model:*

- *Pauli error configurations are sampled from distribution $\mathcal{P}_p$.*
- *Loss configurations are sampled from distribution $\mathcal{P}_l$. However, this loss configuration is not applied directly to the circuit. Instead, we compute the loss-resolving readout $r = \mathcal{R}(l)$ and find the corresponding Pauli envelope $E(r)$.*

The failure probability $P_{\mathrm{fail}}(\mathcal{C}, \mathcal{P}_l, \mathcal{P}_p, \mathsf{Dec})$ is defined as

$$P_{\mathrm{fail}} = \Pr_{\substack{l \sim \mathcal{P}_l \\ p \sim \mathcal{P}_p}} \left[\exists (\mathcal{D}, \mathcal{O}) \in \mathcal{S}(p \oplus E(r), \emptyset) : \mathsf{Dec}(\mathcal{D}, r) \neq \mathcal{O}\right]. \tag{2}$$

Notice that the failure probability does not average over $E(r)$, but counts the entire $E(r)$ as a failure if any error in $E(r)$ causes a decoding failure.

Theorem 1 (Logical Error Rate is Upper-Bounded by Failure Probability)**.**

$$P_{\mathrm{LER}}(\mathcal{C}, \mathcal{P}_l, \mathcal{P}_p, \mathsf{Dec}) \leq P_{\mathrm{fail}}(\mathcal{C}, \mathcal{P}_l, \mathcal{P}_p, \mathsf{Dec}).$$

Proof. See Appendix A 1. □

The key insight is that by Lemma 1, any decoding failure with loss can be traced back to a failure in decoding the Pauli envelope $p \oplus E$ without loss. This establishes that the logical error rate is bounded by the failure probability of decoding the envelope errors.

B. Construction of Pauli Envelope for Atom Loss

The preceding results show that all performance guarantees for the Pauli envelope will automatically transfer to atom loss errors. A natural question arises: can we construct a Pauli envelope with good decoding properties, specifically one that is linear and low-weight? We show below that the answer is affirmative.

Without loss of generality, assume $\mathcal{C}$ is generated by CZ, H, and S gates. For a loss configuration containing a single atom loss location, the Pauli envelope E can be constructed as follows.

Lemma 2 (Pauli Envelope of Atom Loss)**.** *Let $\mathcal{C}$ be a Clifford circuit and l be an atom loss configuration corresponding to a single atom loss location. Then the following construction yields a Pauli envelope E for l.*

Let $\Pi_i = \{I, X, Y, Z\}$ denote the set of all single-qubit Pauli operators at a specific space-time location i. For two such sets, we define $\Pi_i \oplus \Pi_j = \{e_i \oplus e_j : e_i \in \Pi_i, e_j \in \Pi_j\}$ as the set of all pairwise compositions. We identify the relevant locations: (1) the loss location itself, (2) immediately after each subsequent Hadamard gate acting on the lost atom, and (3) just before measuring the lost atom. The Pauli envelope is then $E = \Pi_1 \oplus \Pi_2 \oplus \cdots \oplus \Pi_k$, where $\Pi_1, \ldots, \Pi_k$ are the Pauli sets at these relevant locations.

Proof. See Appendix A 2. □

The key insight of the proof is that a lost atom can be modeled as if reset gates are inserted after each gate following the loss event. Since CZ and S gates preserve $|0\rangle$, redundant resets between consecutive CZ and S gates are not required.

This construction aligns with previous analysis of atom loss [15, 16] when there is only one atom loss. However, this construction naturally generalizes to multiple atom losses through the following linearity properties.

Lemma 3 (Linearity of Pauli Envelope)**.** *(a) Consider a loss configuration $l = l_1 \oplus l_2 \oplus \cdots \oplus l_k$, where each l_i is a single-atom loss configuration. The following composition yields a Pauli envelope for l: $E(l) = \bigoplus_{i=1}^{k} E(l_i)$.*

(b) For a readout r corresponding to a single atom loss with multiple possible loss configurations $\mathcal{L}(r) = \{l^{(1)}, l^{(2)}, \ldots\}$, the following union yields a Pauli envelope for r: $E(r) = \bigcup_{l \in \mathcal{L}(r)} E(l)$.

(c) Let $r^{(m)}$ denote the readout in which only atom m is reported as lost. For a readout r, the following composition over atoms reported as lost yields a Pauli envelope for r: $E(r) = \bigoplus_{m : r_m = 1} E(r^{(m)})$.

Proof. See Appendix A 3. □

This linearity ensures that the Pauli envelope for any combination of atom losses can be constructed from single-atom-loss envelopes, avoiding the need to analyze exponentially many multi-loss scenarios directly or to reason about the nonlinearity of atom loss.

In the following, we mainly use $CNOT$ circuits for simplicity. To apply the results to $CNOT$ circuits, we can view each $CNOT$ as compiled into CZ and H gates. Thus, we need to add Pauli errors whenever a qubit changes its role from a target qubit to a control qubit or vice versa.

We explicitly construct the Pauli envelope for a data atom loss in a standard syndrome extraction circuit for the surface code.

Example 1 (Data qubit loss)**.** *Fig. 4 illustrates a loss configuration with a data-qubit loss. Assume we measure the data qubit after the four CNOTs and it is reported as lost. The loss may have occurred at any of the five locations 1–5, so $\mathcal{L}(r)$ contains five loss configurations. Applying Lemma 2 to each of them and taking the union as in Lemma 3(b), the readout r has Pauli envelope*

$$E = (\Pi_1 \oplus \Pi_2 \oplus \Pi_4 \oplus \Pi_5) \cup (\Pi_2 \oplus \Pi_4 \oplus \Pi_5) \cup (\Pi_3 \oplus \Pi_4 \oplus \Pi_5) \cup (\Pi_4 \oplus \Pi_5) \cup \Pi_5,$$

where each Π_i denotes the set $\{I, X, Y, Z\}$ at the numbered location i in Fig. 4.

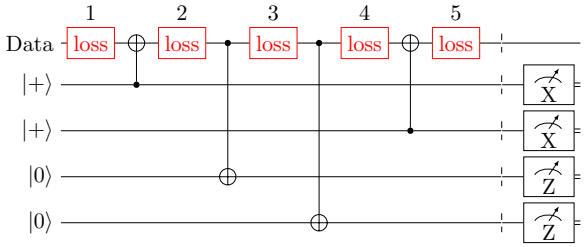


FIG. 4: Surface code syndrome extraction circuit with a data-qubit loss. Red boxes mark the candidate loss locations 1–5.

IV. LOSS-REPLENISHING SYNDROME EXTRACTION FOR SURFACE CODES

As discussed in Section II, specialized syndrome extraction circuits are needed to localize the effects of atom loss in both time and space while satisfying hardware restrictions. In particular, loss-resolving readout can only be applied during measurement, which inevitably collapses the logical information.

In this section, we review the SWAP syndrome extraction, proposed by [34, 35] and adopted by [14, 28, 36], which handles atom loss under these constraints. We then use the Pauli envelope framework to analyze its limitations and introduce a new syndrome extraction circuit, the Mid-SWAP syndrome extraction. With the Envelope-MLE decoder, it doubles the effective distance against atom loss compared to the SWAP syndrome extraction, without additional hardware overhead.

A. SWAP Syndrome Extraction

The SWAP syndrome extraction addresses atom loss on data qubits by exchanging the roles of ancilla and data qubits after each syndrome extraction round. As shown in Fig. 5, the final $CNOT$ interactions (ticks 4 and 8) are reversed, followed by atom shuttling. After the measurements, a classically controlled Pauli gate is applied to the data qubit, which can be implemented in software. This ensures that each atom is measured and replenished within 8 $CNOT$ gates after initialization.

During the data phase (the last 4 ticks), the $CNOT$ direction alternates between control and target. According to Lemma 2, regardless of whether the atom loss occurs during or before the data phase, there will always be data errors on the data qubits.

Furthermore, if atom loss occurs during the ancilla phase, it will create a hook error originating from the ancilla qubit because a Pauli error needs to be inserted at the position of the loss.

Fig. 6 summarizes this observation about the life cycle: an atom loss early in the ancilla phase of the SWAP syndrome extraction leads to **both a hook error and a data error** in the syndrome extraction circuit, making it less robust against atom loss.

In Appendix B 2, we show that this widespread error limits the SWAP syndrome extraction to $d_{\text{loss}} \sim d/2$.

B. Mid-SWAP Syndrome Extraction

To address the possibility of creating both a hook error and a data error from a single atom loss event in the SWAP syndrome extraction, we introduce the Mid-SWAP syndrome extraction, which performs atom shuttling during each syndrome extraction round rather than at the end of each round. This allows each newly replenished atom to serve as a data qubit first and then transition to an ancilla qubit.

As illustrated in Fig. 7, the Mid-SWAP syndrome extraction modifies a standard surface code syndrome extraction circuit by inserting an atom-shuttling step immediately after the first $CNOT$ gate in each round, while all other $CNOT$ gates remain unchanged.

Fig. 8 shows the life cycle of a qubit in the Mid-SWAP syndrome extraction.

If an atom loss occurs during the data phase, it will cause a data error on that data qubit but will not create hook errors during the ancilla phase because the entire ancilla phase is skipped. Moreover, because an ancilla qubit is always a target or always a control of the $CNOT$ gates, Lemma 2 implies that there is no need to insert any Pauli error on the ancilla if it is lost before the beginning of the ancilla phase, except for a possible measurement error.

If an atom loss occurs during the ancilla phase, it will cause a hook error but not a data error because only gates after the loss are affected.

This ensures that in Mid-SWAP syndrome extraction, an atom loss can only cause **either a hook error or a data error**, but not both. Lemma 4 formalizes this observation. Focusing on physical Z errors detected by X-stabilizer measurements and projecting each detector onto the X-Y plane, a Z-ancilla loss in the Mid-SWAP syndrome extraction triggers at most one edge forming a Z-shaped path as illustrated by S_1, S_2, and S_3 in Fig. 9, while an X-ancilla loss triggers at most one edge at a fixed position. This structure is crucial for the decoder analysis in the following sections. After introducing the

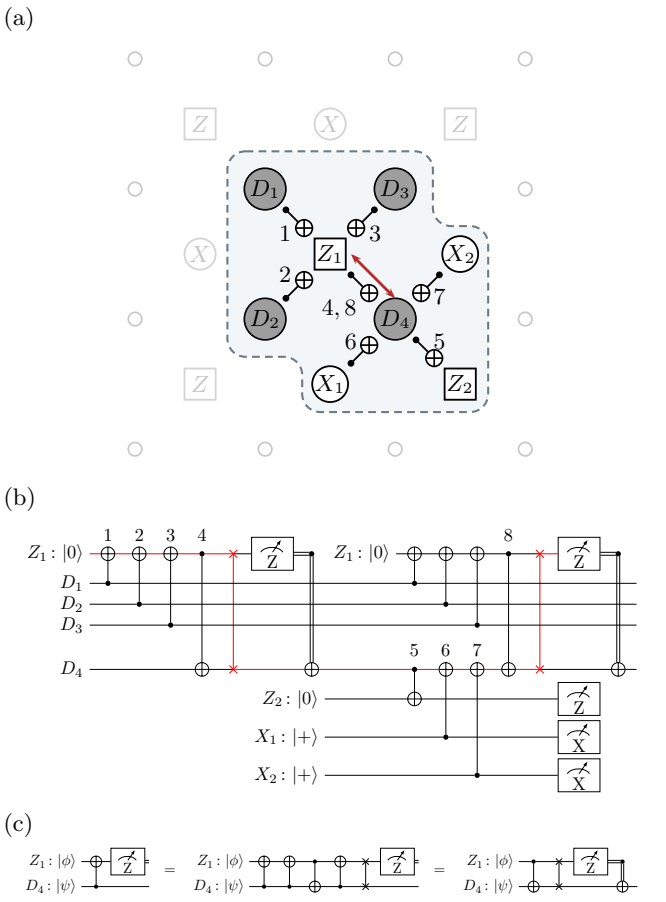


FIG. 5: SWAP syndrome extraction. (a) Spatial layout: $D_1, \ldots, D_4$ are data qubits, and X_1, X_2, Z_1, Z_2 are stabilizer ancillas; names identify the same sites and circuit wires in (a) and (b). Numbers label the tracked atom's interactions over two rounds, and the red double arrow marks atom shuttling after ticks 4 and 8. The dashed boundary selects the qubits shown in (b); $CNOT$ gates involving an outside qubit are omitted. (b) Circuit over two consecutive rounds. The red line tracks one physical atom: it serves as the ancilla Z_1 for ticks 1–4, is shuttled into the data qubit D_4 for ticks 5–8, and is shuttled back and measured after tick 8. The lower three wires are the other stabilizer ancillas adjacent to D_4. (c) Circuit identity validating the swap in (b).

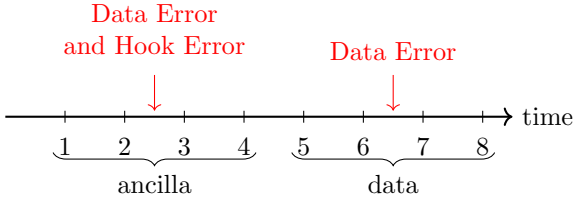


FIG. 6: Timeline of an atom in the SWAP syndrome extraction. The ticks correspond to the gate labels in Fig. 5. The red arrow marks an atom loss between ticks 2 and 3.

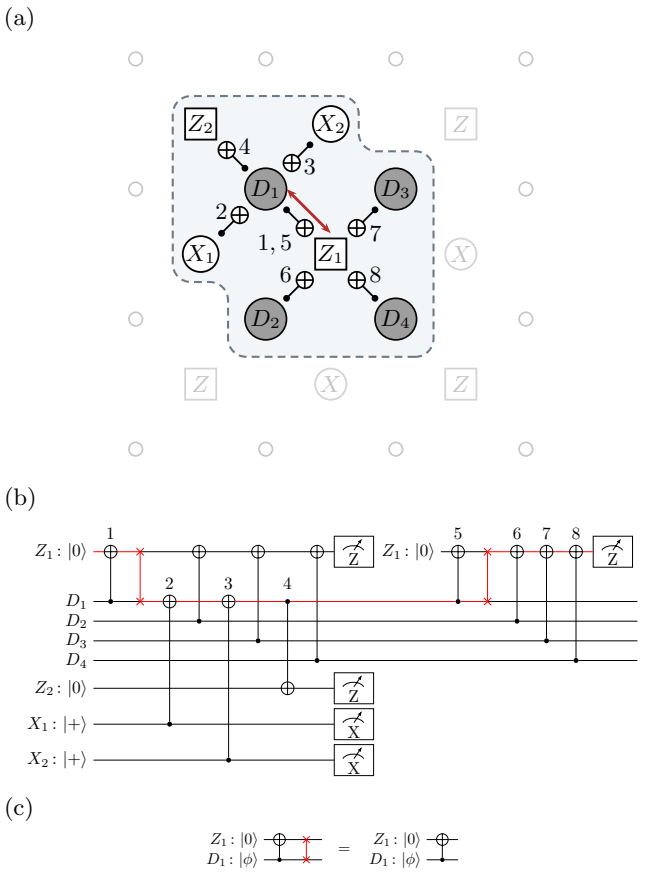


FIG. 7: Mid-SWAP syndrome extraction. Notation and conventions follow Fig. 5. (a) Spatial layout. Atom shuttling (red double arrow) now occurs after ticks 1 and 5. (b) Circuit over two consecutive rounds. The tracked atom (red line) serves as the ancilla Z_1 at tick 1, is shuttled into the data qubit D_1 for ticks 2–4, and is shuttled back after tick 5 to serve as the ancilla for ticks 6–8. (c) Circuit identity validating the swap in (b).

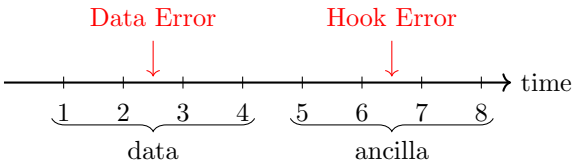


FIG. 8: Timeline of an atom in the Mid-SWAP syndrome extraction. The ticks correspond to the gate labels in Fig. 7.

Envelope-MLE decoder in the next section, we prove the optimality of Mid-SWAP syndrome extraction.

Lemma 4 (Exclusive Loss-induced Patterns in Mid-SWAP syndrome extraction)**.** *Consider a standard surface code with data qubits and ancilla qubits mapped to the grid as shown in Fig. 9. Assume the logical Z observable is defined horizontally, and stabilizers are extracted in the order shown above the figure. We focus on logical Z errors; logical X errors can be handled symmetrically.*

In the matching graph for logical Z decoding, only X-stabilizer measurements (which detect Z-type data errors) are relevant. There are two types of ancilla loss events to consider: loss at a Z-stabilizer ancilla position and loss at an X-stabilizer ancilla position.

For a Z-stabilizer ancilla loss detected at position (x, y), the loss triggers at most one of the following Z-error detector pairs (projected onto the (x, y) plane), which together form a Z-shaped path:

1. $S_1 = ((x-2, y), (x, y+2))$*: a diagonal edge from left to top.*

2. $S_2 = ((x, y+2), (x, y-2))$*: a vertical edge from top to bottom.*

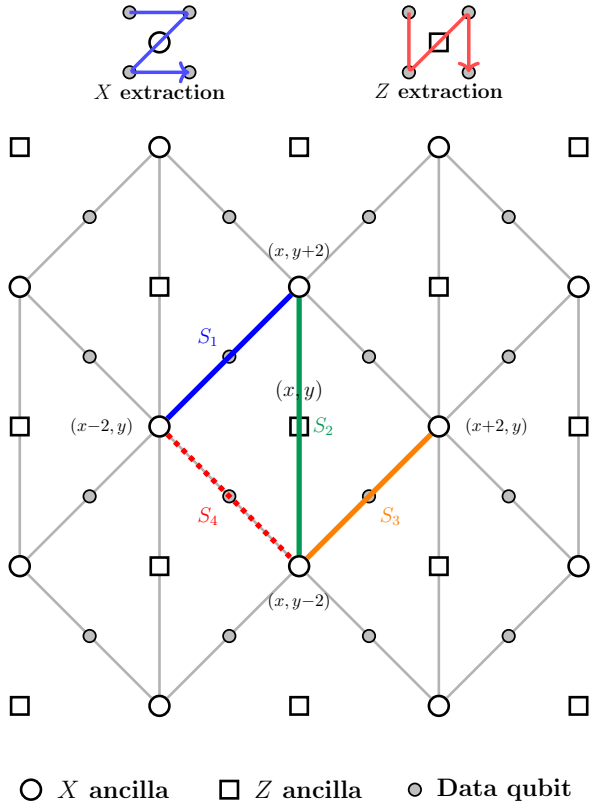


FIG. 9: Detector patterns induced by a single atom loss in the Mid-SWAP syndrome extraction. Each edge represents an error mechanism that can trigger the two endpoint detectors. X-detectors correspond to X stabilizers and are triggered by Z errors; similarly, Z-detectors correspond to Z stabilizers and are triggered by X errors. An atom loss detected at position (x, y) (during Z-ancilla measurements) triggers at most one of three detector pairs: S_1, S_2, and S_3, which together form a Z-shaped path in the matching graph. An atom loss detected at position $(x, y-2)$ (during X-ancilla measurements) can only trigger S_4.

3. $S_3 = ((x, y-2), (x+2, y))$: *a diagonal edge from bottom to right, parallel to* S_1.

For an X-stabilizer ancilla loss detected at position $(x, y-2)$*, the loss triggers* $S_4 = ((x-2, y), (x, y-2))$ *or none of these detector pairs.*

Proof. This follows from enumerating all possible errors; details are provided in Appendix B 1. □

V. ENVELOPE-MLE DECODER

We introduce the Envelope-MLE decoder, formulated as a Mixed-Integer Linear Program (MILP), which jointly optimizes over Pauli envelope selections and Pauli errors to find a minimum-weight error configuration consistent with the observed detector outcomes. Throughout this section, we assume loss-resolving readouts are perfect; extensions to noisy readouts are discussed in Appendix C 5.

Algorithm 1 (Envelope-MLE decoder)**.** *Before receiving the detector patterns and loss-resolving readouts, the decoder first performs the following steps as preprocessing:*

1. ***Find the Pauli envelope for each loss-resolving readout:*** *For each measured atom* m*, find the Pauli envelope* E_m *of the single-loss readout* $r^{(m)}$ *(in which only atom* m *is reported as lost) according to Lemmas 2 and 3.*

2. ***Assign variables for loss patterns:*** *For each pair* $(\mathcal{D}_{m,j}, \mathcal{O}_{m,j}) \in \mathcal{S}(E_m, \emptyset)$*, assign a binary variable* $x_{m,j} \in \{0,1\}$*, indicating whether the loss pattern is selected (*$x_{m,j} = 1$*) or not (*$x_{m,j} = 0$*). Here, the index* m *represents the measured atom, while* j *indexes the distinct detector-observable pairs in* $\mathcal{S}(E_m, \emptyset)$*.*

3. ***Assign variables for Pauli errors:*** *For each Pauli error mechanism* i *in the circuit, assign a binary variable* $y_i \in \{0,1\}$*, indicating whether the Pauli error mechanism is selected (*$y_i = 1$*) or not (*$y_i = 0$*).*

After receiving the detector pattern and loss-resolving readout, the decoder performs the following steps:

1. ***Enforce constraints:***

   (i) Detector satisfaction: *For each detector* u*, enforce*

   $$\sum_i D_{i,u} y_i + \sum_{m,j} D_{m,j,u} x_{m,j} \equiv s_u \pmod 2, \tag{3}$$

   where $D_{i,u} \in \{0,1\}$ *indicates whether Pauli error* i *flips detector* u*,* $D_{m,j,u} \in \{0,1\}$ *indicates whether loss pattern* (m,j) *flips detector* u*, and* $s_u \in \{0,1\}$ *is the observed value of detector* u*.*

   (ii) Loss exclusivity: *For each readout bit* r_m*, where* $r_m = 1$ *indicates that atom* m *is reported as lost, enforce*

   $$\sum_j x_{m,j} = r_m, \tag{4}$$

   ensuring exactly one detector pattern is selected per loss event.

2. ***Minimize the total weight:*** *Subject to the above constraints, solve the following MILP:*

   $$\min_{x,y} \sum_i w_i y_i + \sum_{m,j} \lambda_{m,j} x_{m,j}, \tag{5}$$

   where $w_i = -\log \frac{p_i}{1-p_i}$ *is the log-likelihood ratio of Pauli error* i*, and* $\lambda_{m,j} = -\log \Pr[(\mathcal{D}_{m,j}, \mathcal{O}_{m,j}) \mid r_m = 1]$ *is the negative log-probability of pattern* j *among the patterns of atom* m*, computed from the loss probabilities of the locations in* $\mathcal{L}(r^{(m)})$ *with the Pauli operators of the envelope weighted uniformly.*

3. ***Predict logical observable:*** *Let* $x^*_{m,j}, y^*_i$ *be the optimal solutions to the MILP. Calculate the predicted logical flip:*

   $$\mathcal{O}_{pred} = \bigoplus_i y^*_i \mathcal{O}_i \oplus \bigoplus_{m,j} x^*_{m,j} \mathcal{O}_{m,j} \tag{6}$$

   where $\mathcal{O}_i$ *is the observable flip caused by Pauli error* i*.*

The decoder returns the most likely joint configuration of Pauli errors and envelope patterns in the independent Pauli-envelope model specified by these weights. The envelope guarantees support inclusion, not equality with the physical loss distribution. As is standard for distance proofs [5, 6], the analysis below takes the limit $p \to 0$. If all Pauli error probabilities are proportional to p, then $w_i = \log(1/p) + O(1)$, so for a fixed circuit one additional Pauli error outweighs the bounded envelope-selection weights $\lambda_{m,j}$ and the $O(1)$ differences among the w_i. The decoder then minimizes the number of Pauli errors subject to the envelope constraints, and the proofs set $\lambda_{m,j} = 0$ and $w_i = w$. At practical error rates the bounded terms still affect decoding, so the implementation used in Section VII keeps the full objective.

Lemma 5 (Optimality of Envelope-MLE decoder)**.** *In the limit* $p \to 0$*, the Envelope-MLE decoder finds a solution with Pauli weight no greater than that of the actual error configuration.*

Proof. See Appendix C 1. □

The optimality follows from observing that by Lemma 2, the actual error configuration is always feasible for the MILP. We now show the decoder performance on the Mid-SWAP syndrome extraction, where the localized Pauli envelope structure enables stronger guarantees.

Theorem 2 (MLE Decoder Achieves Optimal Distance)**.** *In the limit $p \to 0$, the Envelope-MLE decoder achieves optimal distance for the Mid-SWAP syndrome extraction. Specifically, the decoder correctly decodes whenever*

$$n_l + 2n_p < d, \tag{7}$$

where n_l is the number of atom losses, n_p is the number of Pauli errors, and d is the code distance.

Proof. See Appendix C 3. □

The proof analyzes the symmetric difference between the actual error and decoder output. By the optimality of the MLE decoder, the Pauli component has weight at most $2n_p w$. Forming an undetectable logical error with n_l loss pattern mismatches requires Pauli weight at least $(d - n_l)w$. Combining these bounds yields the condition $2n_p + n_l \geq d$ for decoder failure.

This bound is tight. When $n_l + 2n_p = d$, consider a length-d logical X string. Suppose the first n_l data qubits on the string are lost, so the remaining segment has length $d - n_l = 2n_p$. Now compare two fault patterns: (i) apply the logical X and add X errors on any n_p of the remaining qubits; (ii) apply no logical operator and instead add X errors on the complementary n_p remaining qubits. After tracing out the lost qubits, the two cases yield the same physical state and identical loss-resolving readouts, so a decoder cannot distinguish them and must fail on at least one, inducing a logical error.

Theorem 2 by itself does not guarantee fault tolerance. We next convert this combinatorial condition into a loss-distance scaling statement for the logical error rate, showing that the logical error rate can be made arbitrarily small by increasing the code distance.

Theorem 3 (Loss Distance of Envelope-MLE decoder)**.** *For the Mid-SWAP syndrome extraction with code distance d, subject to independent atom loss and Pauli errors and decoded by the Envelope-MLE decoder, the loss distance defined in Definition 1 is $d_{\text{loss}} \sim d$.*

Proof. See Appendix C 4. □

The proof follows from standard lattice counting arguments. By defining $\bar{p} := \max(p_{\text{loss}}, \sqrt{p_{\text{pauli}}})$, we unify the error rates and show the sum is dominated by minimum-weight configurations, yielding the stated scaling.

In contrast, the Average-MLE decoder proposed in [14] constructs a detector error model by enumerating all possible detector patterns induced by individual atom losses and combining them with the Pauli detector error model as if all errors are independent. We attribute its suboptimal performance to two limitations that our Envelope-MLE decoder avoids:

- **No exclusivity constraint:** The Average-MLE decoder does not enforce that each atom can be lost at most once. As a result, certain detector patterns caused by Pauli errors may be incorrectly explained as loss errors, even when no actual loss configuration can produce them. This miscorrection limits the decoder to $d_{\text{loss}} \sim d/2$, as we demonstrate in Appendix E 2.
- **Ignoring nonlinear effects:** The Average-MLE decoder enumerates detector patterns from individual losses and combines them linearly. However, multiple simultaneous losses can produce detector patterns that cannot be decomposed into the sum of individual loss patterns. When such nonlinear patterns occur, the decoder must explain them using additional Pauli errors, degrading performance. This effect is particularly significant in the SWAP syndrome extraction as shown in Appendix K.

VI. ENVELOPE-MATCHING DECODER

Matching-based decoders offer efficiency and natural compatibility with transversal logical circuits. Inspired by the exclusivity constraint of the Envelope-MLE decoder, we propose the Envelope-Matching decoder, which achieves $d_{\text{loss}} \sim 2d/3$ for the Mid-SWAP syndrome extraction. This exceeds the $d_{\text{loss}} \sim d/2$ achieved by previous matching-based decoders and is higher than the corresponding Pauli-error exponent $\sim d/2$. Consequently, for comparable small physical error rates, atom-loss errors are asymptotically negligible compared with Pauli errors.

Algorithm 2 (Envelope-Matching Decoder)**.**

1. ***Graph Initialization (for the Z-error sector):*** *Construct a standard matching graph $G = (V, \mathcal{E})$ where vertices V correspond to X-detectors and edges $\mathcal{E}$ correspond to Pauli errors, connecting the detectors that each error flips. Initialize the weight of every edge $e \in \mathcal{E}$ to $w_e = -\log \frac{p_e}{1-p_e}$ where p_e is the probability of this edge being triggered. Let w be the average weight of the edges in the matching graph.*
2. ***Loss Processing:*** *For each atom m with readout bit $r_m = 1$, find the Pauli envelope E_m of $r^{(m)}$ as in Algorithm 1. Then, for each edge $e = (u, v)$, where u and v can be triggered simultaneously by some Pauli errors in E_m, update the weight as follows:*

   Space-like edges *(detectors u and v involving measurements on different ancilla qubits):* $w_e \leftarrow 0.5w$.

   Time-like edges *(detectors u and v only involve measurements on the same ancilla qubit):* $w_e \leftarrow 0.25w$.
3. ***Matching:*** *Run the MWPM decoder on the matching graph G with the updated weights.*

By reweighting the affected edges to weights comparable to regular edges, instead of reducing them

to a constant weight (which is asymptotically negligible compared with Pauli error weights), the Envelope-Matching decoder approximately enforces the exclusivity constraint: selecting multiple edges within the same envelope is discouraged.

Lemma 6 (Failure Condition of Envelope-Matching decoder)**.** *For the Envelope-Matching decoder on the Mid-SWAP syndrome extraction, decoding failure in the presence of n_p Pauli errors and n_l atom losses requires*

$$2n_p + 1.5n_l \geq d. \tag{8}$$

Proof. The proof relies on the fact that after the reweighting, under the equal-edge-weight assumption $w_e = w$ used in Appendix D 1, the reweighted matching graph has logical distance at least $(d - 0.5n_l)w$ for n_l losses, and each loss-induced detector pattern can be matched with total weight at most $0.5w$. See Appendix D 1 for details. □

With this lemma, we can show that the decoder achieves $d_{\text{loss}} \sim 2d/3$ for the Mid-SWAP syndrome extraction.

Theorem 4 (Loss Distance of Envelope-Matching decoder)**.** *For the Mid-SWAP syndrome extraction with code distance d, subject to independent atom loss and Pauli errors, the loss distance is $d_{\text{loss}} \sim 2d/3$.*

Proof. This follows from standard MWPM threshold analysis. See Appendix D 2 for details. □

In contrast, the Marginal-Matching decoder [22, 23] computes the marginal probability that each atom is lost given the loss-resolving readout, adds depolarizing errors with strength proportional to these marginal probabilities, and runs a standard MWPM decoder. Like the Average-MLE decoder, it sets the edges that could be triggered to a constant weight, meaning no exclusivity constraint is enforced, limiting its loss distance to $d_{\text{loss}} \sim d/2$.

VII. EVALUATION

A. Experimental Setup

Error Model. We simulate a noise model combining circuit-level depolarizing noise (probability p_{pauli}) with atom loss (probability p_{loss}). First, we apply standard circuit-level Pauli errors: single-qubit gates and idling locations experience single-qubit depolarizing noise, while two-qubit gates ($CNOT$) experience two-qubit depolarizing noise, all with parameter p_{pauli}. Resets and measurements have bit-flip probability p_{pauli}. Second, atom loss occurs after each operation. For two-qubit gates, loss occurs independently on each qubit with probability $p_{\text{loss}}/2$, so a two-qubit gate causes at least one loss with probability approximately p_{loss}, consistent with previous work [14]. Finally, loss-resolving readouts are also subject to errors: with probability $p_{\text{readout}}/2$, an existing atom is misidentified as lost, and with probability $p_{\text{readout}}/2$, a lost atom is misidentified as existing and assigned a random 0/1 outcome. The distance statements of Sections V and VI assume perfect loss-resolving readouts; here the decoders receive the noisy readouts without modification, which tests their robustness to this error (Appendix C 5 and Theorem 6 give readout-error-aware extensions of the two decoders that we do not use).

We parameterize the noise model using a total error rate p and a loss contribution $\eta \in [0, 1]$, where $p_{\text{readout}} = p_{\text{loss}} = p \cdot \eta$ and $p_{\text{pauli}} = p \cdot (1 - \eta)$.

Since most decoders calculate Pauli weights using $-\log \frac{p_{\text{pauli}}}{1-p_{\text{pauli}}}$, which is undefined when $p_{\text{pauli}} = 0$, we set $p_{\text{pauli}} = 10^{-9}$ for $\eta = 1$ to avoid numerical issues.

Software. We use Stim [37] for circuit simulation and sampling; Appendix M 1 describes how loss configurations are sampled and converted into Stim circuits. For decoding, we employ PyMatching [38] for minimum-weight perfect matching implementations (Envelope-Matching decoder and Marginal-Matching decoder) and Gurobi v13 for MILP-based decoders (Average-MLE decoder and Envelope-MLE decoder), with a 60-second time limit per shot; Appendix M 2 reports more details.

Simulation parameters. We perform $3d$ rounds of syndrome extraction for code distance d to minimize boundary effects. For SWAP syndrome extraction and Mid-SWAP syndrome extraction, we alternate the stabilizer order between rounds: odd rounds use the CNOT order $1, 2, 3, 4$ of Fig. 9 and even rounds the reversed order $4, 3, 2, 1$, because under a fixed order the data qubits along one boundary row and one boundary column would never be replenished; Appendix L explains this and quantifies the effect of the alternation. Hyperparameters for Envelope-Matching decoder are empirically optimized as described in Appendix J.

Data fitting. We extract effective distances using power-law fitting at low physical error rates $P_{\text{LER}} = \alpha(p/p_{\text{th}})^{d_{\text{eff}}}$. Following [14], d_{eff} denotes the fitted exponent: for pure Pauli noise in a fault-tolerant distance-d circuit with odd d, the expected asymptotic exponent is $(d+1)/2$; for pure loss, it depends on the circuit and decoder. We extract thresholds via critical-point fitting following [39] near the threshold regime $P_{\text{LER}} = a + bx + cx^2$ where $x = (p - p_{\text{th}}) \times d^{1/\nu}$. The thresholds in Fig. 10 and those of the matching-based decoders are estimated from $d \in \{5, 7, 9\}$; Appendix M 2 shows the underlying crossing for the Envelope-MLE decoder on Mid-SWAP syndrome extraction.

B. Envelope-MLE Decoder vs Average-MLE Decoder

We compare the Envelope-MLE decoder against the Average-MLE decoder on the SWAP syndrome extrac-

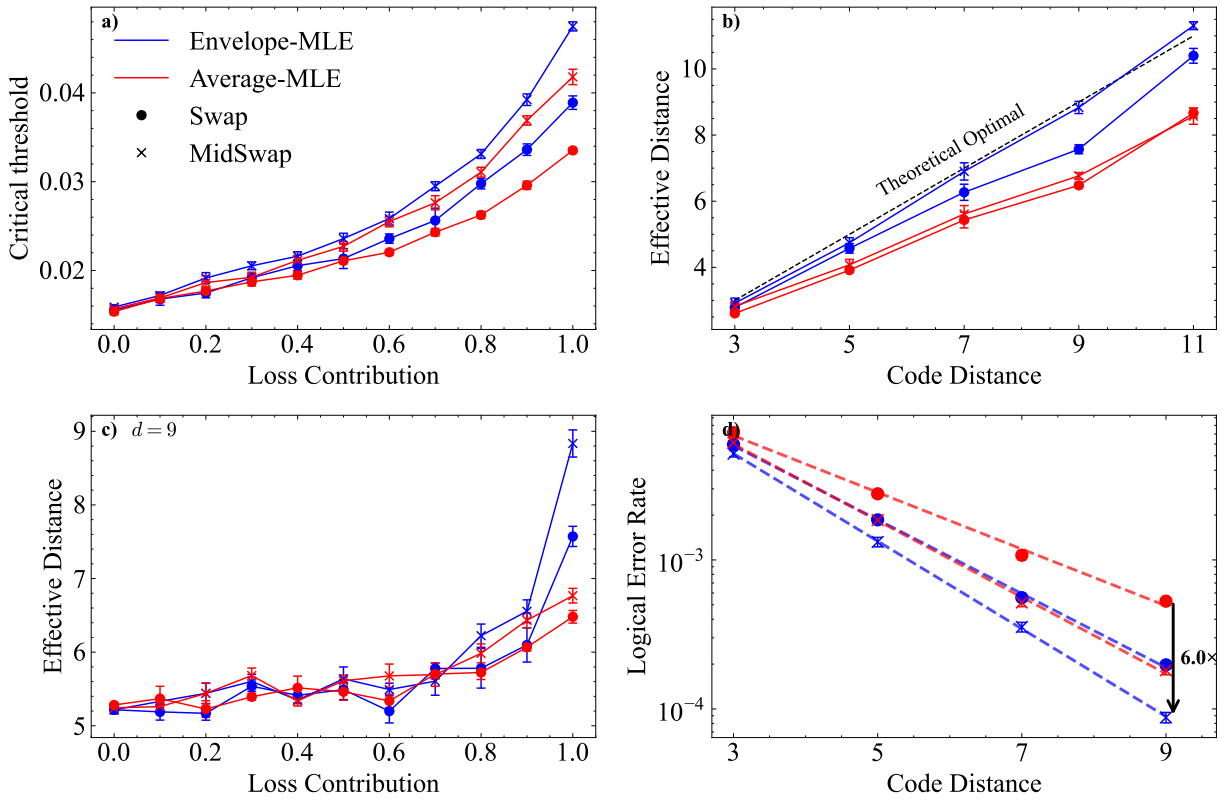


FIG. 10: Comparison of the Envelope-MLE decoder and the Average-MLE decoder on Mid-SWAP syndrome extraction and SWAP syndrome extraction: (a) threshold versus loss contribution η (ratio of atom loss to total error rate p); (b) effective distance versus code distance with $\eta = 1$, up to $d = 11$; (c) effective distance at $d = 9$ versus loss contribution; and (d) distance scaling at $\eta = 90\%$ and total error rate $p = 1.5\%$.

tion and Mid-SWAP syndrome extraction. The teleportation circuit is excluded due to substantially inferior performance (Appendix H).

Fig. 10 shows that Envelope-MLE decoder achieves higher thresholds and effective distances than Average-MLE decoder on both circuits, with Mid-SWAP syndrome extraction outperforming SWAP syndrome extraction in all cases. Combining Envelope-MLE decoder with Mid-SWAP syndrome extraction improves the threshold from 3.35% to 4.75%, a 42% increase over prior work. Changing only the circuit (Average-MLE decoder on Mid-SWAP syndrome extraction) gives 4.18% and changing only the decoder (Envelope-MLE decoder on SWAP syndrome extraction) gives 3.89%, so for the threshold the two contributions are roughly additive. Specifically, at loss contribution $\eta = 1$, Envelope-MLE decoder on Mid-SWAP syndrome extraction achieves $d_{\text{eff}} = 6.9 \pm 0.3$, 8.8 ± 0.2, and 11.3 ± 0.1 at $d = 7$, 9, and 11, matching the theoretical optimum $d_{\text{eff}} = d$. When the total error rate is $p = 1.5\%$ and loss contribution is $\eta = 90\%$, we achieve $6\times$ error suppression over previous work. When extended to correlated atom loss (Appendix G), the Envelope-MLE decoder reveals that correlated loss is *easier* to correct than independent loss: the threshold increases from 5.15% to 7.82% as the correlated fraction grows, because correlated events supply more information about the locations of the lost atoms.

C. Envelope-MLE Decoder on Experimental Data

We evaluate the Envelope-MLE decoder on recent experimental data reported by Ref. [10]. As shown in Fig. 11, replacing the Average-MLE decoder with the Envelope-MLE decoder improves the error suppression factor from $\Lambda = 1.69 \pm 0.08$ to $\Lambda = 2.01 \pm 0.11$. When combined with a machine-learning decoder, the Envelope-MLE decoder further increases Λ from 2.14 ± 0.13 to 2.24 ± 0.14, demonstrating that our framework provides complementary gains on top of learned decoders.

D. Envelope-Matching Decoder vs Marginal-Matching Decoder

We evaluate our scalable Envelope-Matching decoder against the Marginal-Matching decoder on SWAP syndrome extraction and Mid-SWAP syndrome extraction. The Marginal-Matching decoder was originally designed for erasure errors with delayed checks and is therefore

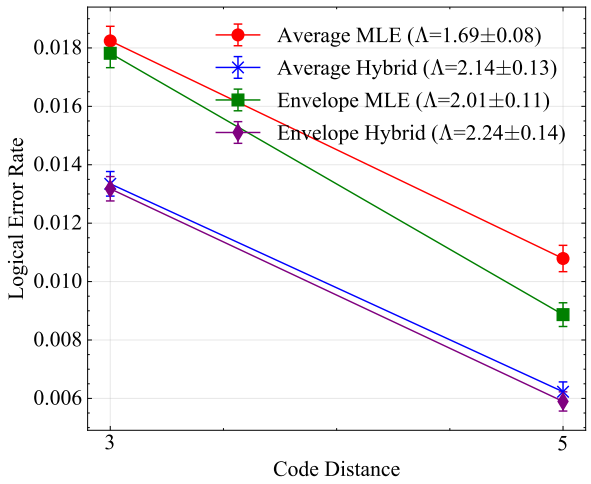


FIG. 11: Per-round logical error rate and error suppression factor Λ ($P_{\mathrm{LER}}^{d=3}/P_{\mathrm{LER}}^{d=5}$) evaluated on experimental data from Ref. [10]. Hybrid refers to combining an MLE decoder with a machine-learning decoder. Replacing the Average-MLE decoder with the Envelope-MLE decoder improves Λ both as a standalone decoder (from 1.69 to 2.01) and when combined with a machine-learning decoder (from 2.14 to 2.24).

not fully aware of the error model of atom loss; nevertheless, we adopt it as our baseline since it represents the best available matching-based decoder from prior work. Concretely, we supply Marginal-Matching decoder with an error model in which each atom loss is replaced by an erasure error with full depolarization, as this is the closest approximation to atom loss within its supported error model. Fig. 12 shows that Envelope-Matching decoder achieves substantially higher thresholds and effective distances on both circuits, with Mid-SWAP syndrome extraction consistently outperforming SWAP syndrome extraction. Combining Envelope-Matching decoder with Mid-SWAP syndrome extraction improves the threshold from 3.42% to 4.38%, a 30% increase over prior work.

Marginal-Matching decoder exhibits severe performance degradation as $\eta \to 1$: as $p_{\mathrm{pauli}} \to 0$, the Pauli weights $-\log \frac{p_{\mathrm{pauli}}}{1-p_{\mathrm{pauli}}} \to \infty$, while loss-induced edges retain constant weight, causing the decoder to over-select loss-induced edges. Theoretical analysis confirms that Marginal-Matching decoder is limited to $d_{\mathrm{loss}} \sim d/2$ (Appendix E 2), whereas our approach achieves $d_{\mathrm{loss}} \sim 2d/3$.

E. Compatibility with Transversal Logical Circuits

Beyond memory experiments, we evaluate Envelope-Matching decoder on transversal logical circuits. We consider a two-logical-qubit circuit that repeatedly performs transversal $CNOT$ gates (with the first qubit as control and the second as target) after each syndrome extraction cycle, followed by a logical Z measurement on the target qubit. Fig. 13 shows that Envelope-Matching decoder with Mid-SWAP syndrome extraction achieves higher thresholds and effective distances than Marginal-Matching decoder, demonstrating that our approach naturally extends to logical circuit decoding.

VIII. SUMMARY AND OUTLOOK

We have presented the Pauli envelope, a theoretical framework for analyzing atom loss in neutral-atom quantum computers. Using this framework, we designed the Mid-SWAP syndrome extraction, a syndrome extraction circuit that achieves an optimal loss distance $d_{\mathrm{loss}} \sim d$ with the Envelope-MLE decoder, and $d_{\mathrm{loss}} \sim 2d/3$ with the efficient Envelope-Matching decoder.

Our results demonstrate that atom loss, despite requiring destructive detection, can achieve the same loss distance as erasure errors and a much higher threshold than Pauli errors. This shows that atom loss need not limit the achievable distance in the delayed-detection, gate-removal model studied here and provides new insights for future hardware and decoder co-design. As shown in Appendix G, correlated atom loss is *easier* to correct than independent loss, with the threshold increasing from 5.15% to 7.82% as the correlated fraction grows when only considering atom loss on two-qubit gates. This suggests that correlated atom-loss events should not pose an additional challenge in this noise model, though a more detailed analysis under realistic hardware noise models is needed.

Several directions remain open for future work. First, extending our methods beyond surface codes remains open; we include a brief discussion on general CSS codes in Appendix F. Second, the performance gap between Envelope-MLE decoder and Envelope-Matching decoder suggests room for intermediate decoders that balance optimality and efficiency. Third, dual-rail superconducting qubits exhibit a similar error model to atom loss. Applying our framework to this platform is a promising direction. Finally, the Pauli envelope can serve as an input feature for machine learning-based decoders like those in [40], potentially improving their interpretability and generalization.

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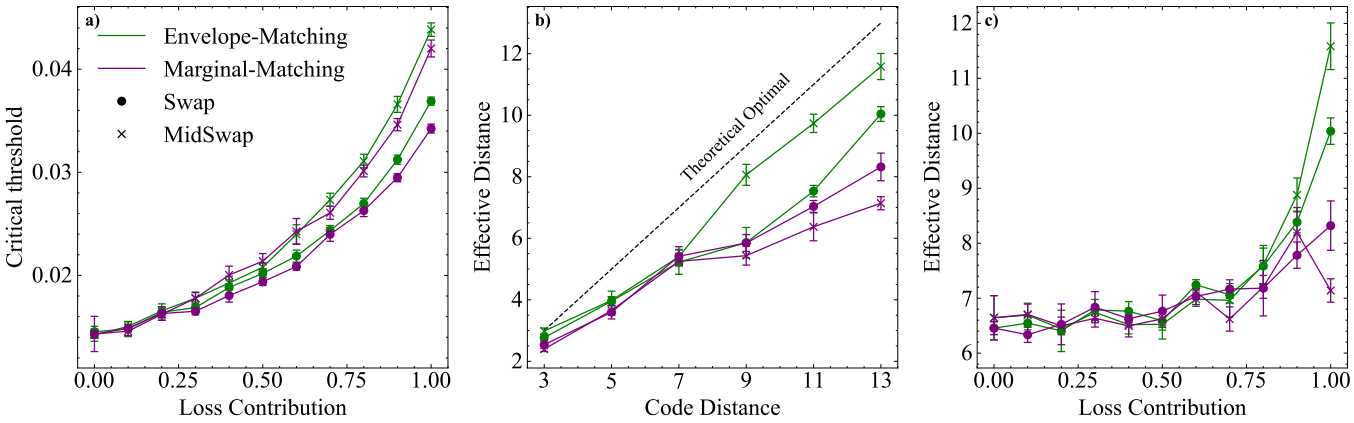


FIG. 12: Comparison of the Envelope-Matching decoder and the Marginal-Matching decoder on Mid-SWAP syndrome extraction and SWAP syndrome extraction: (a) threshold versus loss contribution; (b) effective distance versus code distance with $\eta = 1$; (c) effective distance at $d = 13$ versus loss contribution.

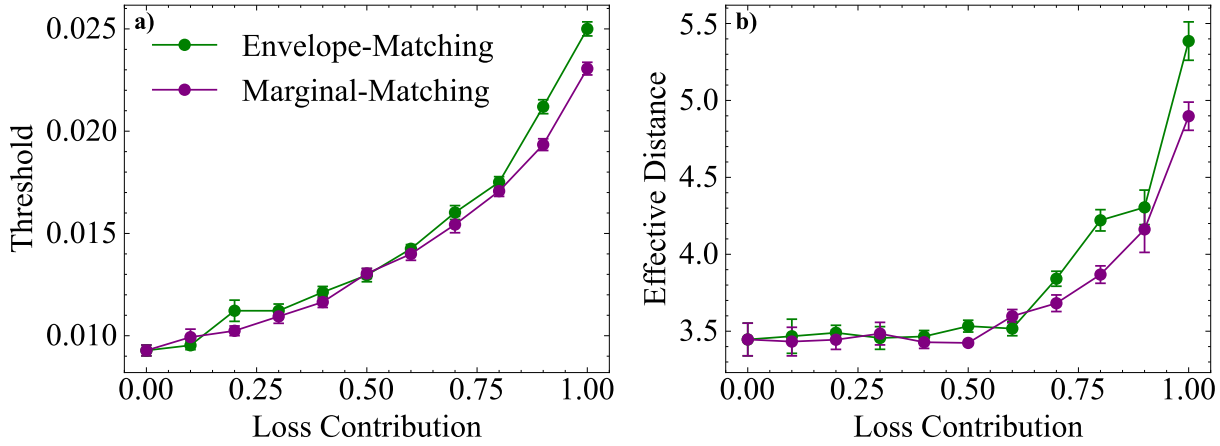


FIG. 13: Comparison of the Envelope-Matching decoder and the Marginal-Matching decoder for transversal logical circuits with Mid-SWAP syndrome extraction: (a) threshold versus loss contribution; (b) effective distance at $d = 7$ versus loss contribution.

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Appendix A: Proofs from Pauli Envelope Section

This section contains the detailed proofs of lemmas and theorems presented in the Pauli envelope section.

1. Proof of Effective Distance Theorem

Theorem 1 (Logical Error Rate is Upper-Bounded by Failure Probability)**.**

$$P_{\mathrm{LER}}(\mathcal{C}, \mathcal{P}_l, \mathcal{P}_p, \mathit{Dec}) \leq P_{\mathrm{fail}}(\mathcal{C}, \mathcal{P}_l, \mathcal{P}_p, \mathit{Dec}).$$

Proof. Consider any loss configuration l with readout $r = \mathcal{R}(l)$ and Pauli error p. Let E denote the Pauli envelope of r. By Lemma 1, if $\mathsf{Dec}(\cdot, r)$ correctly decodes all detector-observable pairs in $\mathcal{S}(p \oplus E, \emptyset)$, then it also correctly decodes all pairs in $\mathcal{S}(p, \mathcal{L}(r))$.

Taking the contrapositive: for any realized detector-observable pair at fixed (p, l) where the decoder fails, i.e., $\mathsf{Dec}(\mathcal{D}(p,l), \mathcal{R}(l)) \neq \mathcal{O}(p,l)$, there must exist some $(\mathcal{D}, \mathcal{O}) \in \mathcal{S}(p \oplus E, \emptyset)$ such that $\mathsf{Dec}(\mathcal{D}, \mathcal{R}(l)) \neq \mathcal{O}$.

Averaging over circuit measurements, assigned bits, and the sampled loss and Pauli configurations, we obtain

$$\begin{aligned}
&P_{\mathrm{LER}}(\mathcal{C}, \mathcal{P}_l, \mathcal{P}_p, \mathsf{Dec}) \\
&= \Pr_{\substack{l \sim \mathcal{P}_l \\ p \sim \mathcal{P}_p}} [\mathsf{Dec}(\mathcal{D}(p,l), \mathcal{R}(l)) \neq \mathcal{O}(p,l)] \\
&\leq \Pr_{\substack{l \sim \mathcal{P}_l \\ p \sim \mathcal{P}_p}} [\exists (\mathcal{D}, \mathcal{O}) \in \mathcal{S}(p \oplus E, \emptyset) : \mathsf{Dec}(\mathcal{D}, \mathcal{R}(l)) \neq \mathcal{O}] \\
&= P_{\mathrm{fail}}(\mathcal{C}, \mathcal{P}_l, \mathcal{P}_p, \mathsf{Dec}).
\end{aligned}$$

□

2. Proof of Pauli Envelope of Atom Loss

Lemma 2 (Pauli Envelope of Atom Loss)**.** *Let $\mathcal{C}$ be a Clifford circuit and l be an atom loss configuration corresponding to a single atom loss location. Then the following construction yields a Pauli envelope E for l.*

Let $\Pi_i = \{I, X, Y, Z\}$ denote the set of all single-qubit Pauli operators at a specific space-time location i. For two such sets, we define $\Pi_i \oplus \Pi_j = \{e_i \oplus e_j : e_i \in \Pi_i, e_j \in \Pi_j\}$ as the set of all pairwise compositions. We identify the relevant locations: (1) the loss location itself, (2) immediately after each subsequent Hadamard gate acting on the lost atom, and (3) just before measuring the lost atom. The Pauli envelope is then $E = \Pi_1 \oplus \Pi_2 \oplus \cdots \oplus \Pi_k$, where $\Pi_1, \ldots, \Pi_k$ are the Pauli sets at these relevant locations.

Proof. We construct a sequence of equivalent circuits, illustrated in Fig. 14.

Step 1 ($\mathcal{C} \to \mathcal{C}_1$): Insert a reset gate after every gate following the loss event. Since the lost atom is in the $|0\rangle$ state, all the CZ gates have no effect.

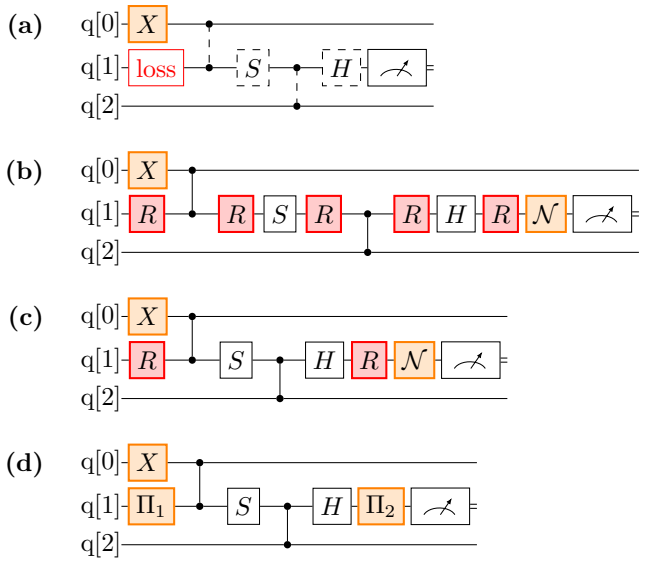


FIG. 14: Construction of the Pauli envelope for atom loss. The orange X gate represents a Pauli error p. (a) Original circuit $\mathcal{C}$ with a loss event (red box) on qubit q_1. (b) Circuit $\mathcal{C}_1$ where reset gates R are inserted after every gate following the loss, and a completely depolarizing channel $\mathcal{N}$ is inserted before the measurement. (c) Circuit $\mathcal{C}_2$ where resets are only at key locations: the loss location and after each Hadamard gate; $\mathcal{N}$ is kept. CZ and S gates preserve $|0\rangle$, so intermediate resets are redundant. (d) Circuit $\mathcal{C}_3$, which is equivalent to $\mathcal{C}$ with Pauli envelope $\Pi_1 \oplus \Pi_2$, where Π_i is the Pauli set at labeled location i.

Furthermore, a completely depolarizing channel $\mathcal{N}$ is added before the measurement to model the random bit b_m assigned to the lost atom.

We use $\mathcal{S}(\mathcal{C})$ to denote the set of possible outcomes of $\mathcal{C}$. By our modeling of atom loss, $\mathcal{S}(\mathcal{C}) = \mathcal{S}(\mathcal{C}_1)$.

Step 2 ($\mathcal{C}_1 \to \mathcal{C}_2$): Retain resets only at two key locations: (1) the loss location, and (2) immediately after each subsequent Hadamard gate on the lost atom; the depolarizing channel $\mathcal{N}$ before the measurement is kept. Since CZ and S gates preserve the $|0\rangle$ state, the intermediate resets are redundant, giving $\mathcal{S}(\mathcal{C}_1) = \mathcal{S}(\mathcal{C}_2)$.

Step 3 ($\mathcal{C}_2 \to$ Pauli envelope): Let E denote all possible Pauli errors at these three types of locations (the two reset locations and the location of $\mathcal{N}$), and let $\mathcal{C}_3$ be $\mathcal{C}_2$ with each reset and $\mathcal{N}$ replaced by the Pauli set Π at that location. We claim $\mathcal{S}(\mathcal{C}_2) \subseteq \mathcal{S}(\mathcal{C}_3)$. This is a statement about supports only; no probabilities of the two circuits are compared. A reset is not a mixture of Pauli operators, but the completely depolarizing channel is, and it contains the reset: for any state ρ of the circuit, a reset of qubit q gives $R(\rho) = |0\rangle\langle 0|_q \otimes \mathrm{Tr}_q \rho$, whereas

$$\mathcal{N}(\rho) = \tfrac{1}{2}\mathbb{1}_q \otimes \mathrm{Tr}_q \rho = \tfrac{1}{2}R(\rho) + \tfrac{1}{2}X_q R(\rho) X_q.$$

Hence every outcome that has nonzero probability after R also has nonzero probability after $\mathcal{N}$. Since $\mathcal{N}(\rho) = \frac{1}{4}\sum_{P\in\Pi} P\rho P$, every outcome that has nonzero probability after $\mathcal{N}$ has nonzero probability for at least one Pauli operator $P \in \Pi$ inserted at that location. Applying this at each of the three types of locations gives $\mathcal{S}(\mathcal{C}_2) \subseteq \mathcal{S}(\mathcal{C}_3)$.

Notice $\mathcal{C}_3$ is just $\mathcal{C}$ with Pauli envelope E, and none of the above derivations depends on whether there are additional Pauli errors. Combining these steps, we have $\forall p, \mathcal{S}(p, l) \subseteq \mathcal{S}(p \oplus E, \emptyset)$. □

3. Proof of Pauli Envelope for Multiple Losses

Lemma 3 (Linearity of Pauli Envelope)**.** *(a) Consider a loss configuration $l = l_1 \oplus l_2 \oplus \cdots \oplus l_k$, where each l_i is a single-atom loss configuration. The following composition yields a Pauli envelope for l: $E(l) = \bigoplus_{i=1}^{k} E(l_i)$.*

(b) For a readout r corresponding to a single atom loss with multiple possible loss configurations $\mathcal{L}(r) = \{l^{(1)}, l^{(2)}, \ldots\}$, the following union yields a Pauli envelope for r: $E(r) = \bigcup_{l\in\mathcal{L}(r)} E(l)$.

(c) Let $r^{(m)}$ denote the readout in which only atom m is reported as lost. For a readout r, the following composition over atoms reported as lost yields a Pauli envelope for r: $E(r) = \bigoplus_{m:r_m=1} E(r^{(m)})$.

Proof. **Part (a):** Consider a loss configuration $l = l_1 \oplus l_2 \oplus \cdots \oplus l_k$ with k loss locations. We construct a sequence of circuits $\mathcal{C}_0, \mathcal{C}_1, \cdots, \mathcal{C}_k$, where $\mathcal{C}_i$ is obtained from $\mathcal{C}_{i-1}$ by removing the gates affected by loss l_i, thus $\mathcal{S}_{\mathcal{C}_i}(p, \emptyset) = \mathcal{S}_{\mathcal{C}_{i-1}}(p, l_i)$. Let $E(l_i)$ denote the Pauli envelope for loss l_i. Applying Lemma 2 to the transition from $\mathcal{C}_{i-1}$ to $\mathcal{C}_i$, we have $\mathcal{S}_{\mathcal{C}_i}(p, \emptyset) = \mathcal{S}_{\mathcal{C}_{i-1}}(p, l_i) \subseteq \mathcal{S}_{\mathcal{C}_{i-1}}(p \oplus E(l_i), \emptyset)$. Iterating this relation backwards from k to 0, we obtain

$$\begin{aligned}
\mathcal{S}_{\mathcal{C}}(p, l) &= \mathcal{S}_{\mathcal{C}_k}(p, \emptyset) \\
&\subseteq \mathcal{S}_{\mathcal{C}_{k-1}}(p \oplus E(l_k), \emptyset) \\
&\subseteq \mathcal{S}_{\mathcal{C}_{k-2}}(p \oplus E(l_k) \oplus E(l_{k-1}), \emptyset) \\
&\vdots \\
&\subseteq \mathcal{S}_{\mathcal{C}_0}\left(p \oplus \bigoplus_{i=1}^{k} E(l_i), \emptyset\right).
\end{aligned}$$

Thus $E(l) = \bigoplus_{i=1}^{k} E(l_i)$ is a valid Pauli envelope.

Part (b): For a readout r with possible loss configurations $\mathcal{L}(r) = \{l^{(1)}, l^{(2)}, \ldots\}$, we have $\mathcal{S}(p, \mathcal{L}(r)) = \bigcup_{l\in\mathcal{L}(r)} \mathcal{S}(p, l)$. By definition, $\mathcal{S}(p, l) \subseteq \mathcal{S}(p \oplus E(l), \emptyset)$, so $\mathcal{S}(p, \mathcal{L}(r)) \subseteq \bigcup_l \mathcal{S}(p \oplus E(l), \emptyset) = \mathcal{S}(p \oplus \bigcup_l E(l), \emptyset)$. Thus $E(r) = \bigcup_{l\in\mathcal{L}(r)} E(l)$.

Part (c): Let $M_r = \{m : r_m = 1\}$. The possible configurations are $\mathcal{L}(r) = \{\bigoplus_{m\in M_r} l_m : l_m \in \mathcal{L}(r^{(m)})\}$. By part (a), $E(\bigoplus_m l_m) = \bigoplus_m E(l_m)$. By part (b), $E(r) = \bigcup_{(l_m)_m} \bigoplus_m E(l_m) = \bigoplus_m \bigcup_{l_m} E(l_m) = \bigoplus_m E(r^{(m)})$, with all indices $m \in M_r$, where the second equality uses the distributive property $(A \cup B) \oplus (C \cup D) = (A \oplus C) \cup (A \oplus D) \cup (B \oplus C) \cup (B \oplus D)$. □

Appendix B: Detector Patterns from a Single Atom Loss

In this section, we illustrate how each atom loss event triggers a set of detector flips. For simplicity, we consider the Z-error sector, detected by X-stabilizer measurements, as these errors are already sufficient for our proofs. The detector patterns for both Mid-SWAP syndrome extraction and SWAP syndrome extraction are shown in Fig. 15.

1. Mid-SWAP Syndrome Extraction

The detector patterns for Mid-SWAP syndrome extraction are shown in Fig. 15a and Fig. 15c when a loss-resolving readout is triggered on Z- and X-type ancilla qubits, respectively. To generate these figures, we first convert each atom loss event to its Pauli envelope, then identify all affected edges in the matching graph. These results will be sufficient for the proof of Lemma 4.

2. SWAP Syndrome Extraction

For SWAP syndrome extraction, the detector patterns are shown in Fig. 15b and Fig. 15d when a loss-resolving readout is triggered on Z- and X-type ancilla qubits, respectively. In this case, we do not convert the atom loss event to its Pauli envelope, but instead directly identify all affected edges by removing the gates that act on the lost atom. This approach is used because these figures will be employed to prove the upper bound on the loss distance in the next section, where we need to ensure that the failure events can actually occur due to atom loss.

Appendix C: Proofs from Envelope-MLE decoder Section

This section contains the detailed proofs of lemmas and theorems from the Envelope-MLE decoder section, in the limit $p \to 0$ of Section V, i.e., with $\lambda_{m,j} = 0$ and $w_i = w$.

1. Proof of Optimality of Envelope-MLE decoder

Lemma 5 (Optimality of Envelope-MLE decoder)**.** *In the limit $p \to 0$, the Envelope-MLE decoder finds a solution with Pauli weight no greater than that of the actual error configuration.*

Proof. According to Lemma 2, the Pauli envelope E constructed for each loss configuration satisfies the property that any detector-observable pair produced by the circuit with loss can also be produced by the circuit with errors from E and no loss.

Therefore, the actual error configuration (consisting of actual Pauli errors and loss events with their corresponding Pauli envelopes) is a feasible solution to the MILP.

Since the MILP minimizes the total Pauli error weight over all feasible solutions, and the actual error is feasible, the optimal solution found by the decoder must have weight no greater than the weight of the actual error. □

2. Proof of Loss Pattern Weight Requirement

We work in one error sector of the matching graph. For each loss-induced pair $(\mathcal{D}_{m,j}, \mathcal{O}_{m,j})$, choose a representative error chain $F_{m,j}$ with detector boundary $\partial F_{m,j} = \mathcal{D}_{m,j}$ and logical observable flip $\mathcal{O}_{m,j}$. Here $\oplus$ on chains denotes symmetric difference.

For n_l atoms m_i reported as lost, let j_i and k_i index the pairs in a Pauli-envelope representation of the actual error and in the decoder's solution, respectively. Their loss component is the chain

$$C_L = \bigoplus_{i=1}^{n_l} (F_{m_i,j_i} \oplus F_{m_i,k_i}),$$

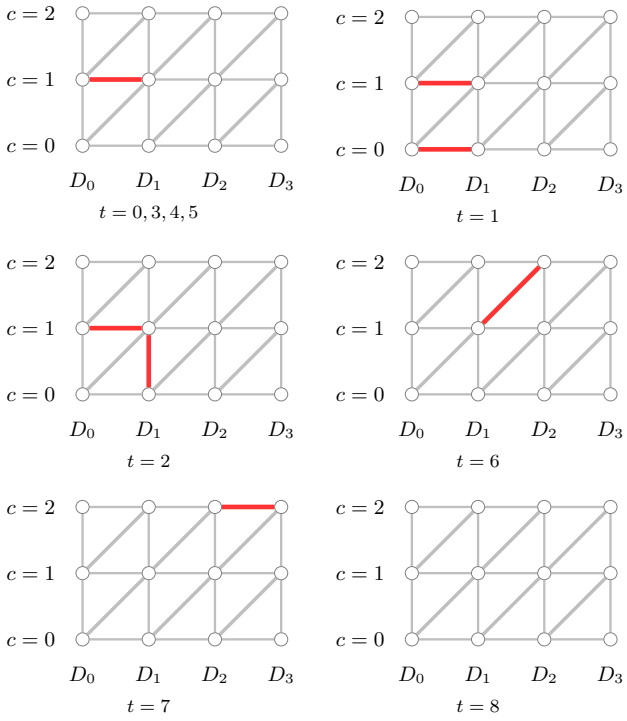
whose boundary is the syndrome difference $\partial C_L = \bigoplus_{i=1}^{n_l} (\mathcal{D}_{m_i,j_i} \oplus \mathcal{D}_{m_i,k_i})$. Let C_P be the symmetric difference of the actual and decoded Pauli-error chains. Then $C = C_P \oplus C_L$ has empty detector boundary. If the decoder fails, C is an *undetectable logical error*: a chain with empty detector boundary and a nontrivial logical observable flip.

Lemma 7. *For the Mid-SWAP syndrome extraction with distance d, if $C_P \oplus C_L$ forms an undetectable logical error, then $|C_P| \geq d - n_l$, where n_l is the number of loss events, and $|\cdot|$ denotes the number of errors, or the weight divided by w.*

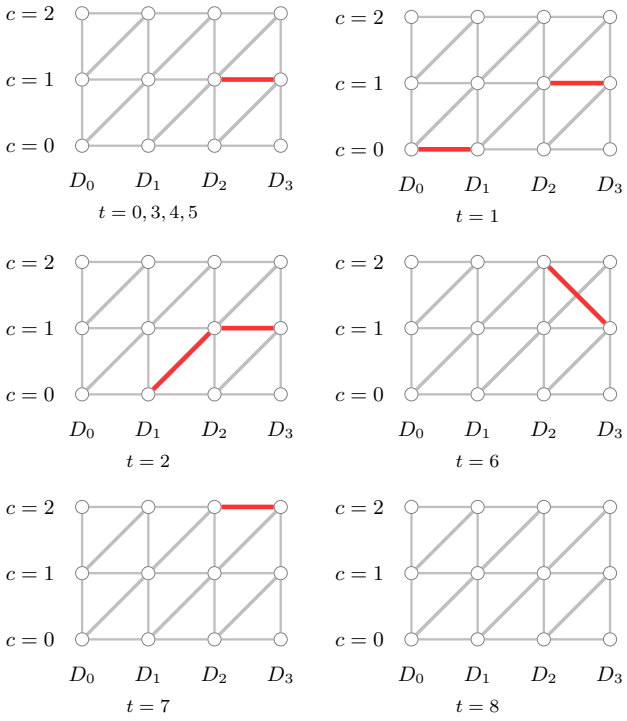
Proof. Work in the Z-error sector; the X sector is analogous. We fix one logical path with at least d right-going edges and show that each Pauli mechanism and each loss mismatch covers at most one of the right-going edges.

Projection onto space. Use the code-capacity matching graph: its vertices are the X-checks, and each data qubit is an edge joining the checks that detect a Z error on it; a data qubit on the left or right boundary has only one such check, and its edge ends on that boundary. The four data qubits of a bulk Z-stabilizer bound a diamond-shaped face. This graph has no hook or time-like edges, and a hook such as S_2 becomes two data-qubit edges (Fig. 16(a,b)). Propagating a fault through the ideal circuit to the data qubits at the end of the circuit gives its support, a set of edges defined up to Z-stabilizers. Each Pauli mechanism in C_P and each loss mismatch $F_{m_i,j_i} \oplus F_{m_i,k_i}$ in C_L has a support, and their symmetric difference is the projection of C.

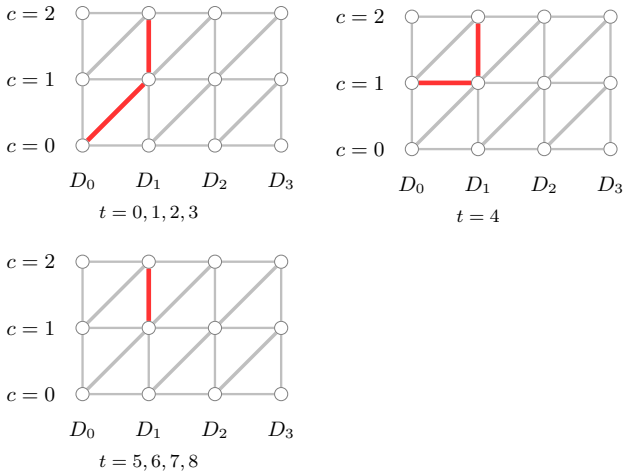
Local support shapes. Label the data qubits of a bulk Z-stabilizer face $q_{\mathrm{NW}}, q_{\mathrm{SW}}, q_{\mathrm{NE}}, q_{\mathrm{SE}}$ (Fig. 16(b)); the Z-extraction order of Fig. 9 is $q_{\mathrm{NW}}, q_{\mathrm{SW}}, q_{\mathrm{NE}}, q_{\mathrm{SE}}$. A



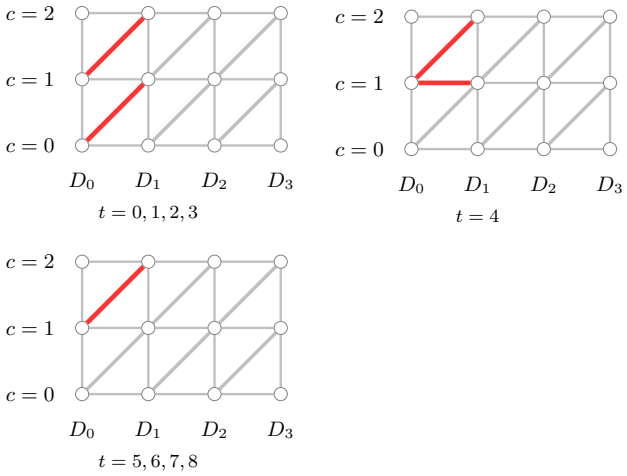
(a) Mid-SWAP syndrome extraction (Z-type ancilla loss)



(b) SWAP syndrome extraction (Z-type ancilla loss)



(c) Mid-SWAP syndrome extraction (X-type ancilla loss)



(d) SWAP syndrome extraction (X-type ancilla loss)

FIG. 15: Loss-induced detector patterns for different surface code variants. Each red edge can be triggered independently and will flip its two endpoints. This is a flattened view of the S_1–S_2–S_3 path as shown in Fig. 9. t represents the time step (the number of $CNOT$ gates since the last reset) when the atom loss event occurs. c represents the index of syndrome extraction rounds.

Pauli fault on the Z-ancilla or one of its CNOTs propagates to a suffix of this order, which up to the face stabilizer is empty, a single edge, or the vertical hook $\{q_{\mathrm{NW}}, q_{\mathrm{SW}}\}$ (equivalently $\{q_{\mathrm{NE}}, q_{\mathrm{SE}}\}$). For a loss at this ancilla, Lemma 4 gives the patterns $\emptyset, S_1, S_2, S_3$ with supports

$$S_1 : \{q_{\mathrm{NW}}\}, \qquad S_2 : \{q_{\mathrm{NW}}, q_{\mathrm{SW}}\}, \qquad S_3 : \{q_{\mathrm{SE}}\}.$$

Equal patterns cancel, and the other mismatches are

- a single edge: S_1, S_3, $S_1 \oplus S_2 = \{q_{\mathrm{SW}}\}$, and $S_2 \oplus S_3 = \{q_{\mathrm{NW}}, q_{\mathrm{SW}}, q_{\mathrm{SE}}\}$, which is equivalent to $\{q_{\mathrm{NE}}\}$ up to stabilizers;
- the vertical hook: S_2;
- two opposite edges: $S_1 \oplus S_3 = \{q_{\mathrm{NW}}, q_{\mathrm{SE}}\}$.

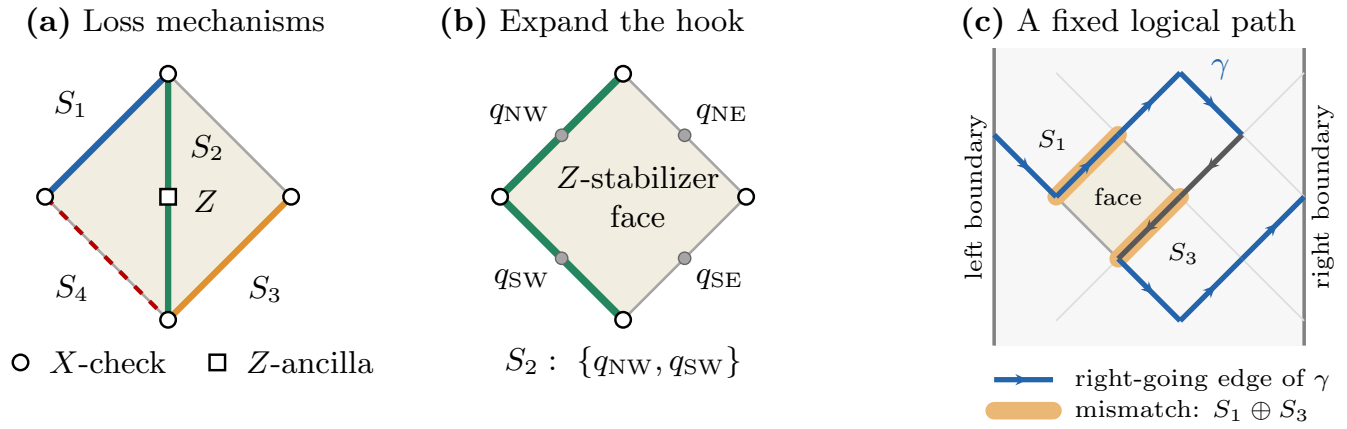


FIG. 16: Spatial supports and path counting for Lemma 7 in the Z-error sector. (a) The local loss mechanisms from Fig. 9; circles denote X-checks and the square denotes the Z-ancilla. (b) In the code-capacity matching graph, each side of the diamond is one data qubit. The hook S_2 expands into the two green edges; adding the face stabilizer exchanges them for the two right-side edges. NW, SW, NE, and SE denote the four compass positions. (c) A simple path γ joins the left and right boundaries. Blue arrows indicate its right-going edges and gray arrows its left-going edges. The orange bands mark the mismatch $S_1 \oplus S_3$. Although the path uses both marked edges, it traverses them in opposite horizontal directions, so the mismatch covers only one right-going edge.

A loss at an X-ancilla gives only the fixed edge S_4 or nothing, and at a weight-two boundary Z-stabilizer every support reduces to at most one edge. Every support is therefore empty, a single edge, a vertical hook, or two opposite edges of one face.

A left-to-right path. Since C has empty detector boundary, its projection has even degree at every X-check, so it decomposes into cycles and paths ending on the boundary. Cycles and paths returning to the boundary they started from are products of Z-stabilizers; since C flips the logical observable, an odd number of the paths join the left boundary to the right one. Choose one, remove any loops so that it visits no vertex twice, and orient the resulting path γ from left to right. Each edge changes the data-qubit column by one and the boundaries are d columns apart, so γ has at least d right-going edges.

Every support contains at most one right-going edge of γ; only the two-edge shapes need an argument. The two edges of a vertical hook meet at the west (or east) corner of the face, so if both were right-going, γ would leave (or enter) that corner twice. For two opposite edges, close γ by an arc outside the code patch. By the Jordan curve theorem [42], the resulting simple closed polygon separates the plane into two regions, one on the left and one on the right of γ all along it. The interior of the face is connected and disjoint from the polygon, so the face lies on the same side of every edge of γ on its boundary. Traversed rightward, however, q_{NW} has the face on its right and q_{SE} on its left, so at most one of them is right-going (Fig. 16(c)).

Counting. Each loss event contributes at most one right-going edge of γ, and so does each Pauli mechanism in C_P. Since every right-going edge of γ lies in the projection of C, it is contributed by at least one of them. Hence $d \le |C_P| + n_l$. □

3. Proof of Envelope-MLE decoder Achieves Optimal Distance

Theorem 2 (MLE Decoder Achieves Optimal Distance)**.** *In the limit $p \to 0$, the Envelope-MLE decoder achieves optimal distance for the Mid-SWAP syndrome extraction. Specifically, the decoder correctly decodes whenever*

$$n_l + 2n_p < d, \tag{7}$$

where n_l is the number of atom losses, n_p is the number of Pauli errors, and d is the code distance.

Proof. Suppose the actual error pattern consists of n_l atom losses and n_p Pauli errors.

Use the decomposition $C = C_P \oplus C_L$ from Appendix C 2, where C_P is the Pauli chain difference and C_L is the difference of the selected loss representative chains.

By the optimality of the Envelope-MLE decoder (Lemma 5), the decoder outputs Pauli weight at most $n_p w$. Thus, the Pauli component satisfies:

$$|C_P| \le 2n_p. \tag{C1}$$

For decoder failure, C must be a non-trivial undetectable logical error. Applying Lemma 7 with the n_l loss pattern mismatches in C_L, we obtain $|C_P| \ge d - n_l$, and therefore:

$$|C_P| \ge d - n_l. \tag{C2}$$

Combining these bounds, decoder failure requires:

$$2n_p + n_l \geq d. \tag{C3}$$

Therefore, the decoder succeeds when $2n_p + n_l < d$. This bound is tight: there exists a configuration of d losses (e.g., forming a chain across the lattice) that completely erases the logical information, making recovery impossible for any decoder. This proves the optimality of both the Envelope-MLE decoder and the Mid-SWAP syndrome extraction. □

4. Proof of Loss-Distance Scaling for Envelope-MLE decoder

Theorem 3 (Loss Distance of Envelope-MLE decoder)**.** *For the Mid-SWAP syndrome extraction with code distance d, subject to independent atom loss and Pauli errors and decoded by the Envelope-MLE decoder, the loss distance defined in Definition 1 is $d_{\text{loss}} \sim d$.*

Proof. By Theorem 2, decoder failure requires an undetectable logical error with n_l losses and n_p Pauli errors satisfying $\omega := 2n_p + n_l \geq d$.

We bound the failure probability by summing over all connected error configurations with effective weight $\omega \geq d$. Let $N(s)$ denote the number of connected subgraphs of size s in the lattice. Since each error triggers a constant number of detectors, let μ be the maximum number of errors that can trigger overlapping detectors with a fixed error, which is also a constant. We have $N(s) \leq C(d)\mu^s$ for a polynomial function $C(d)$ and a constant μ [43].

For a configuration with s errors containing k losses and $s-k$ Pauli errors, the effective weight is $\omega = k + 2(s-k) = 2s - k$, occurring with probability $p_{\text{loss}}^k p_{\text{pauli}}^{s-k}$.

Define $\bar{p} := \max(p_{\text{loss}}, \sqrt{p_{\text{pauli}}})$. Then:

$$p_{\text{loss}}^k p_{\text{pauli}}^{s-k} \leq \bar{p}^k (\bar{p}^2)^{s-k} = \bar{p}^{2s-k} = \bar{p}^\omega.$$

Summing over configurations:

$$P_{\text{fail}} \leq \sum_{\omega=d}^{\infty} \sum_{s=\lceil \omega/2 \rceil}^{\omega} C(d)\mu^s \binom{s}{2s-\omega} \bar{p}^\omega \tag{C4}$$

$$\leq \sum_{\omega=d}^{\infty} \tilde{C}(d)\tilde{\mu}^\omega \bar{p}^\omega, \tag{C5}$$

where $\tilde{C}(d), \tilde{\mu}$ absorb the inner sum and account for the combinatorial factors.

For $\bar{p} < p_{\text{th}} := 1/\tilde{\mu}$, this geometric series converges to the following:

$$P_{\text{fail}} \leq \frac{\tilde{C}(d)(\tilde{\mu}\bar{p})^d}{1 - \tilde{\mu}\bar{p}} \tag{C6}$$

$$= \frac{\tilde{C}(d)}{1 - \tilde{\mu}\bar{p}} \left(\frac{\bar{p}}{p_{\text{th}}} \right)^d \tag{C7}$$

$$\leq \frac{\tilde{C}(d)}{1 - \tilde{\mu}\bar{p}} \left(\frac{p_{\text{loss}}^d + p_{\text{pauli}}^{d/2}}{p_{\text{th}}^d} \right) \tag{C8}$$

$$= O\left(p_{\text{th}}^{-d} (p_{\text{loss}}^d + p_{\text{pauli}}^{d/2}) \right). \tag{C9}$$

The third step follows from $\bar{p} = \max(p_{\text{loss}}, \sqrt{p_{\text{pauli}}})$, which gives $\bar{p}^d = \max(p_{\text{loss}}^d, p_{\text{pauli}}^{d/2})$. □

5. Extension to Loss-Resolving Errors

Theorem 5 (Extension to Loss-Resolving Errors)**.** *The Envelope-MLE decoder can be extended to handle loss-resolving errors. Suppose with probability $p_{\text{readout}} = p_{\text{loss}}$ an atom is detected as lost but is actually not lost, and with probability p_{loss} a lost atom is randomly assigned to 0 or 1 with equal probability. We modify the Envelope-MLE decoder as follows: instead of enforcing $\sum_j x_{m,j} = r_m$ when $r_m = 0$, we relax this constraint to $\sum_j x_{m,j} \leq 1$ and add a penalty term $w \sum_j x_{m,j}$ to the objective function. This modification ensures the same failure probability scaling as without loss-resolving errors:*

$$P_{\text{fail}} = O\left(p_{\text{th}}^{-d} (p_{\text{loss}}^d + p_{\text{pauli}}^{d/2}) \right).$$

Proof. Loss-resolving errors fall into two categories:

False positives (loss detected but atom not actually lost): These do not require special treatment. When the decoder is told an atom was lost (but it wasn't), we can treat the atom as if it were lost just before measurement, and this case is already handled by the original decoder.

False negatives (atom lost but not detected): For false negatives, the key observation is that an undetected atom loss requires two independent events: (1) the atom is lost, and (2) the loss-resolving measurement fails to detect it. Therefore, the effective probability is p_{loss}^2.

Notice that after adding the penalty term, an undetected atom loss has the same form as a Pauli error; by repeating the proof of Lemma 7, we can show that the failure condition generalizes to $2n_p + n_l + 2n_f \geq d$, where n_f is the number of false negatives.

Following the proof structure of Theorem 3, we can show that the failure probability scaling is unchanged:

$$P_{\text{fail}} = O\left(p_{\text{th}}^{-d} (p_{\text{loss}}^d + p_{\text{pauli}}^{d/2}) \right).$$

□

We did not use this extension in our numerical evaluation as it significantly increases MILP problem sizes.

Appendix D: Proofs from Envelope-Matching decoder Section

This section contains the detailed proofs of lemmas and theorems from the Envelope-Matching decoder section. For simplicity, we assume all baseline Pauli-edge weights satisfy $w_e = w$.

1. Proof of Failure Condition

We first establish the following property regarding the matching graph weights. Recall that in Algorithm 2, we set each edge weight to w, and then for each edge affected by atom loss, we reduce its weight to $0.5w$ for space-like edges and $0.25w$ for time-like edges.

Lemma 8. *For the Mid-SWAP syndrome extraction and any configuration of n_l atom losses, the reweighted matching graph in Algorithm 2 has the following properties:*

1. *The detector pattern induced by each loss can be matched with total weight at most $0.5w$.*
2. *Every undetectable logical error chain C satisfies $\mathrm{Weight}(C) \geq (d - 0.5n_l)w$.*

Proof. The first property follows from Figs. 15a and 15c; the only case worth noting is $t = 1$ in Fig. 15a, where four detectors can be triggered and the pattern is matched with two weight-$0.25w$ time-like edges.

For the second property, fix the full loss configuration and an undetectable logical error chain C. Work in the Z-error sector; the X sector is analogous. We group the space-like edges of C and show that at least d groups are needed. Each group has weight w, except for at most one group of weight $0.5w$ per loss.

Spatial supports. As in the proof of Lemma 7, propagate errors to the data qubits at the end of the circuit and represent their supports in the code-capacity matching graph. Time-like edges are measurement-bit flips and have empty support. A Pauli mechanism, a single loss pattern, or the symmetric difference of two patterns from the same loss has, up to stabilizers, one of four support shapes: the empty set, a single edge, a vertical hook, or two opposite edges of one stabilizer face.

Each such support contains at most one right-going edge of any simple left-to-right logical path. For a vertical hook, traversing both edges rightward would require the path to leave or enter the same vertex twice. For two opposite edges, traversing both rightward would put the face interior on opposite sides of the path. This is impossible as discussed in the proof of Lemma 7.

Grouping the edges. Assign each weight-$0.5w$ space-like edge of C to one atom reported as lost whose readout reweights it. If several atoms qualify, choose any one, so that each edge is assigned exactly once. For each atom, pair its assigned edges arbitrarily, leaving one unpaired edge if their number is odd. This partitions all space-like edges of C into three types of groups:

1. a groups consisting of one edge of weight w each;
2. b groups consisting of two weight-$0.5w$ edges assigned to the same atom, with total weight w per group;
3. c groups consisting of one unpaired weight-$0.5w$ edge each.

Each atom leaves at most one unpaired edge, so

$$c \leq n_l.$$

A group of the second type has the support of the symmetric difference of two patterns from the same loss; a group of the third type has the support of a single pattern. Thus every group admits one of the support shapes described above.

Counting groups. Choose such a support representative for each group. Their symmetric difference agrees with the spatial projection of C up to stabilizers, so it still represents a nontrivial logical error. It therefore contains a simple path γ joining the left and right boundaries. Since these boundaries are d columns apart and each spatial edge changes the column by one, γ has at least d right-going edges. Each group contains at most one of them, while every edge of γ belongs to at least one group. Consequently,

$$a + b + c \geq d.$$

Overlaps between group supports do not affect this count: an edge surviving their symmetric difference must occur in at least one support.

Bounding the weight. The space-like edges have total weight $(a + b + 0.5c)w$. Dropping the nonnegative weight of the time-like edges gives

$$\begin{aligned}\mathrm{Weight}(C) &\geq (a + b + 0.5c)w \\ &= (a + b + c - 0.5c)w \\ &\geq (d - 0.5c)w \\ &\geq (d - 0.5n_l)w.\end{aligned}$$

This proves the second property. □

Lemma 6 (Failure Condition of Envelope-Matching decoder)**.** *For the Envelope-Matching decoder on the Mid-SWAP syndrome extraction, decoding failure in the presence of n_p Pauli errors and n_l atom losses requires*

$$2n_p + 1.5n_l \geq d. \tag{8}$$

Proof. Let F denote the actual error as a set of edges in the matching graph. The MWPM decoder predicts an error correction $\widehat{F}$ such that $\mathrm{Weight}(\widehat{F}) \leq \mathrm{Weight}(F)$. The decoder fails when $F \oplus \widehat{F}$ forms an undetectable logical error.

Using Lemma 8, we derive the bounds:

1. By Property 1, each loss event contributes weight at most $0.5w$ to the error matching. Thus, for n_p Pauli errors and n_l losses:

$$\text{Weight}(F) \leq (n_p + 0.5n_l)w. \tag{D1}$$

2. Apply Property 2 directly to the undetectable logical chain $F \oplus \widehat{F}$. Since $\text{Weight}(F \oplus \widehat{F}) \leq \text{Weight}(F) + \text{Weight}(\widehat{F})$, we have:

$$\text{Weight}(F) + \text{Weight}(\widehat{F}) \geq (d - 0.5n_l)w. \tag{D2}$$

Combining these bounds with the failure condition $2\,\text{Weight}(F) \geq \text{Weight}(F) + \text{Weight}(\widehat{F})$:

$$2(n_p + 0.5n_l)w \geq (d - 0.5n_l)w. \tag{D3}$$

Simplifying yields the final condition:

$$2n_p + 1.5n_l \geq d. \tag{D4}$$

□

2. Proof of Loss-Distance Scaling for Envelope-Matching decoder

Theorem 4 (Loss Distance of Envelope-Matching decoder)**.** *For the Mid-SWAP syndrome extraction with code distance d, subject to independent atom loss and Pauli errors, the loss distance is $d_{\text{loss}} \sim 2d/3$.*

Proof. Similar to the proof of Theorem 3, we define the effective weight $\omega = 2n_p + 1.5n_l$. For an error configuration with n_p Pauli errors and n_l losses, the probability is $p_{\text{pauli}}^{n_p} p_{\text{loss}}^{n_l}$.

Define $\bar{p} := \max(p_{\text{loss}}^{2/3}, \sqrt{p_{\text{pauli}}})$. Then:

$$p_{\text{pauli}}^{n_p} p_{\text{loss}}^{n_l} \leq (\bar{p}^2)^{n_p} (\bar{p}^{1.5})^{n_l} = \bar{p}^{2n_p + 1.5n_l} = \bar{p}^{\omega}. \tag{D5}$$

Following the same counting argument as in Theorem 3, we sum over all error configurations with weight $\omega \geq d$:

$$P_{\text{fail}} \leq \sum_{\omega=d}^{\infty} \tilde{C}(d) \tilde{\mu}^{\omega} \bar{p}^{\omega}. \tag{D6}$$

For $\bar{p} < p_{\text{th}} := 1/\tilde{\mu}$, the series converges and is dominated by the $\omega = d$ term:

$$P_{\text{fail}} = O(p_{\text{th}}^{-d} \bar{p}^{d}) = O\left(p_{\text{th}}^{-d} (p_{\text{loss}}^{2d/3} + p_{\text{pauli}}^{d/2})\right). \tag{D7}$$

□

3. Extension to Loss-Resolving Errors

Theorem 6 (Extension to Loss-Resolving Errors)**.** *The decoder can be extended to handle loss-resolving errors by initializing time-like edges with weight $w_T = 0.5w$ instead of w. The decoder maintains:*

$$P_{\text{fail}} = O\left(p_{\text{th}}^{-d} (p_{\text{loss}}^{2d/3} + p_{\text{pauli}}^{d/2})\right).$$

Proof. The extension handles two types of loss-resolving errors:

False positives: Similar to the proof of Theorem 5, this case is already handled by the original decoder.

False negatives (atom lost but not detected): With the modified initialization ($w_T = 0.5w$), an undetected loss induces a detector pattern. The key observation is:

- The total weight for matching this pattern is at most w, equivalent to a single Pauli error.
- With no atom lost, the weight of a logical error is still dw.
- With n_l detected losses, every undetectable logical error has weight at least $(d - 0.5n_l)w$.

Therefore, in the failure condition analysis, an undetected loss (n_f false negatives) contributes similarly to Pauli errors:

$$2(n_p + n_f) + 1.5n_l \geq d. \tag{D8}$$

Since false negatives happen with probability p_{loss}^2, and it requires $d/2$ false negatives to form a logical error, repeating the proof of Theorem 3, we can show that this contributes a term that scales like $(p_{\text{loss}}^2)^{d/2}$, which is dominated by $p_{\text{loss}}^{2d/3}$.

Therefore, the failure probability remains:

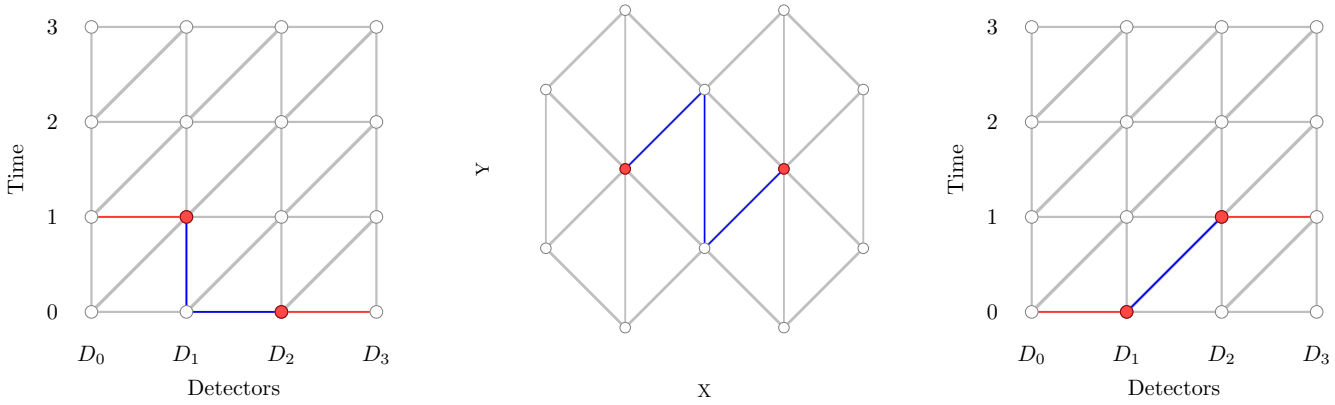
$$P_{\text{fail}} = O\left(p_{\text{th}}^{-d} (p_{\text{loss}}^{2d/3} + p_{\text{pauli}}^{d/2})\right).$$

□

We did not use this extension in our numerical evaluation as it significantly reduces the threshold for Pauli errors.

Appendix E: Upper Bound on the Loss Distance

We analyze the upper bound on the loss distance for (1) the conventional matching decoder, (2) Average-MLE decoder and Marginal-Matching decoder, and (3) SWAP syndrome extraction by directly constructing undecodable error patterns.



(a) A space-time slice of the matching graph showing a scenario where the conventional matching decoder fails.

(b) A space-space slice of the matching graph showing a scenario where the Average-MLE decoder and Marginal-Matching decoder fail.

(c) A space-time slice of the matching graph showing a scenario where the SWAP syndrome extraction fails.

FIG. 17: Failure scenarios for (a) the conventional matching decoder, (b) the Average-MLE decoder and Marginal-Matching decoder, and (c) the SWAP syndrome extraction. Red dots represent detector flips. Red and blue edges represent two sets of edges with (asymptotically) equal weight that can produce the same detector flips, which cannot be distinguished by the decoder.

1. Conventional matching decoder

A conventional matching decoder decodes as if only Pauli errors were present. However, this approach only achieves $d_{\text{loss}} \sim d/3$.

As shown in Fig. 17a, if errors similar to those in Fig. 15a ($t = 2$) occur, the conventional decoder incorrectly interprets this pattern as a weight-2 error, leading to miscorrection.

If $d/3$ such errors occur, the decoder fails.

2. Average-MLE decoder and Marginal-Matching decoder

The Average-MLE decoder [14] constructs a detector error model by enumerating all possible single-loss patterns and combining them with the Pauli detector error model. However, this approach only achieves $d_{\text{loss}} \sim d/2$, as shown in Fig. 17b.

When an ancilla qubit is lost, the Average-MLE decoder will reweight all the blue edges to a constant weight, while other edges scale like $-\log \frac{p_{\text{pauli}}}{1-p_{\text{pauli}}}$.

It is possible that the ancilla loss triggers no detector at all (this can happen with a constant probability). However, since the blue path has a weight that becomes negligible compared with other edges, the decoder will connect the two red detectors using this path, even when the detectors are actually caused by boundary Pauli errors.

If $d/2$ such errors occur, the decoder fails.

3. SWAP syndrome extraction

As shown in Fig. 17c, when a qubit is lost, either the red paths or the blue path can produce the same detector flips, and these cases have the same probability. Therefore, there is no way to distinguish them.

If $d/2$ such errors occur, any decoder fails. These cases can actually happen as illustrated in Fig. 15b ($t = 1$ and $t = 2$).

Appendix F: Extension to Other CSS Codes

In this section, we discuss how to extend our analysis beyond surface codes to more general CSS codes.

To make the Mid-SWAP technique applicable, we assume the following structure for the CSS code. We believe these assumptions can be relaxed by using different alternating syndrome extraction orders in time.

Definition 5 (Structure of a CSS Code)**.** *Let C be a CSS code with parameters $[[n, k, d]]$. Assume there are n_x X-stabilizers and n_z Z-stabilizers such that $n_x + n_z = n$. (In general, these stabilizers need not be independent; we assume this so that the Mid-SWAP technique can be applied.)*

Let each X-type stabilizer S_i^X act on data qubits $q_{i,1}^X, q_{i,2}^X, \ldots, q_{i,m_i}^X$, and let each Z-type stabilizer S_j^Z act on data qubits $q_{j,1}^Z, q_{j,2}^Z, \ldots, q_{j,m_j}^Z$.

We assume that all $q_{i,1}^X$ for all $i \in \{1, \ldots, n_x\}$ and $q_{j,1}^Z$ for all $j \in \{1, \ldots, n_z\}$ are distinct. In other words, there is a one-to-one correspondence between ancilla qubits and data qubits such that each data qubit appears as the first qubit in exactly one stabilizer.

When the context is clear, we drop the superscripts X and Z for simplicity.

Definition 6 (Hook error)**.** *For each X-stabilizer S_i acting on data qubits $q_{i,1}, q_{i,2}, \ldots, q_{i,m}$, we define a hook error $H_{i,t}$ as follows:*

$$H_{i,t} = \prod_{j=1}^{t} X_{q_{i,j}}. \tag{F1}$$

This describes how an X error on an X-ancilla a_i at time step t propagates to the data qubits. Note that the hook error $H_{i,t}$ has weight at most $m/2$ since we can always apply S_i to reduce its weight to an equivalent Pauli error.

We define the hook-error distance d_h of $\mathcal{C}$ as the minimum number of hook errors and regular Pauli errors combined needed to implement a nontrivial logical operator.

Formally, we have

$$(\prod_{i=1}^{n_h} H_{i,t_i}) \times (\prod_{i=1}^{n_p} X_{q_i}) \tag{F2}$$

where any such product with $n_h + n_p < d_h$ does not implement a nontrivial logical operator. The hook-error distance for logical Z operators is defined similarly.

For any CSS code, we can construct a simple syndrome extraction circuit that uses only X-type and Z-type ancilla qubits to extract the syndrome with a circuit-level distance of d_h for Pauli errors.

Example 2. *A syndrome extraction circuit can be constructed using the following steps:*

1. *Assign n_x X-type ancilla qubits to the X-type stabilizers, and n_z Z-type ancilla qubits to the Z-type stabilizers.*
2. *For each syndrome extraction cycle, we initialize X-type ancilla qubits in the $|+\rangle$ state and Z-type ancilla qubits in the $|0\rangle$ state.*
3. *For each X-type ancilla a_i associated with qubits $q_{i,1}, q_{i,2}, \ldots, q_{i,m}$, apply $CNOT(a_i, q_{i,1})$, $CNOT(a_i, q_{i,2})$, …, $CNOT(a_i, q_{i,m})$.*
4. *For each Z-type ancilla a_i associated with qubits $q_{i,1}, q_{i,2}, \ldots, q_{i,m}$, apply $CNOT(q_{i,1}, a_i)$, $CNOT(q_{i,2}, a_i)$, …, $CNOT(q_{i,m}, a_i)$.*
5. *Measure all the X-type ancilla qubits in the X-basis and all the Z-type ancilla qubits in the Z-basis.*

Some examples of codes satisfying the condition $d_h = d$ include surface codes, hypergraph product (HGP) codes [44], and some bivariate bicycle (BB) codes [45]. For such codes, the syndrome extraction circuits are also optimal in terms of circuit-level distance [46, 47].

We can apply the Mid-SWAP technique to this syndrome extraction circuit to obtain a new syndrome extraction circuit with atom replenishment.

Example 3 (Mid-SWAP syndrome extraction circuit)**.** *Formally, for the third and fourth steps of Example 2, we insert a $SWAP$ gate between the ancilla-data qubit pair after the first $CNOT$ gate.*

1. *For each X-type ancilla a_i associated with qubits $q_{i,1}, q_{i,2}, \ldots, q_{i,m}$, apply $CNOT(a_i, q_{i,1})$, $SWAP(a_i, q_{i,1})$, $CNOT(a_i, q_{i,2})$, …, $CNOT(a_i, q_{i,m})$.*
2. *For each Z-type ancilla a_i associated with qubits $q_{i,1}, q_{i,2}, \ldots, q_{i,m}$, apply $CNOT(q_{i,1}, a_i)$, $SWAP(q_{i,1}, a_i)$, $CNOT(q_{i,2}, a_i)$, …, $CNOT(q_{i,m}, a_i)$.*

To analyze the tolerance to atom loss, we need to generalize the loss pattern of Lemma 4 to this CSS setting.

Lemma 9 (Loss-induced hook error)**.** *Let $\mathcal{C}$ be the Mid-SWAP syndrome extraction circuit from Example 3.*

Let a_i be an X-type ancilla associated with qubits $q_{i,1}, q_{i,2}, \ldots, q_{i,m}$. If an atom loss of a_i is detected, then the Pauli envelope has the following form when propagated to the end of this syndrome extraction cycle:

$$E = \{I_{q_{i,1}}, Z_{q_{i,1}}\} \oplus \{H_{i,t}\}_{t=0}^{m} \oplus \{\textit{Measurement errors}\}. \tag{F3}$$

where $H_{i,t}$ is the hook error

$$H_{i,t} = \prod_{j=1}^{t} X_{q_{i,j}}. \tag{F4}$$

Measurement errors, which flip the $0/1$ outcome of the ancilla readout, project to the empty set on the data qubits and are ignored below. They are distinct from loss-resolving readout errors, which misreport whether an atom was lost (Section VII).

Furthermore, if a_i is a Z-type ancilla, it will only trigger a data error on $q_{i,1}$.

Proof. The construction of the Mid-SWAP syndrome extraction circuit ensures that each atom loss affects at most two consecutive syndrome extraction cycles. Thus, we only need to analyze two consecutive cycles and will focus on the remaining errors on the data qubits after these two cycles. Suppose the atom loss occurs at time step t (after the t-th $CNOT$ gate) and is detected on the ancilla qubit a_i at the end of the second cycle.

1. $0 \le t \le m$: the atom is lost during the data phase. The Pauli envelope can be written as

$$E_i = \bigoplus_{\tau=t}^{m} \Pi_{q_{i,1}}^{\tau} \oplus \Pi_{a_i}^{m+1}, \tag{F5}$$

where $\Pi_q^\tau = \{I_q^\tau, X_q^\tau, Y_q^\tau, Z_q^\tau\}$ denotes the Pauli error set on qubit q at time step τ.

For any element in the first term of the Pauli envelope $\bigoplus_{\tau=t}^{m} \Pi_{q_{i,1}}^{\tau}$, the residual error on the data qubit is an arbitrary Pauli error on $q_{i,1}$.

For the second term of the Pauli envelope $\Pi_{a_i}^{m+1}$, this corresponds to an error on ancilla qubit a_i after the first $CNOT$ gate in the second cycle; when propagated to the data qubits, it becomes an X error or identity.

Therefore, the joint effect of these two terms is an arbitrary Pauli error on data qubit $q_{i,1}$, which can be written in the form

$$E = \{I_{q_{i,1}}, Z_{q_{i,1}}\} \oplus \{H_{i,t}\}_{t=0}^{1}. \tag{F6}$$

2. $m < t \leq 2m$: The Pauli envelope contains only a single Pauli operator:

$$E_i = \Pi_{a_i}^{t}. \tag{F7}$$

When this error propagates to the data qubits, it becomes an X-type hook error of the form

$$H_{i,t-m} = \prod_{j=1}^{t-m} X_{q_{i,j}}. \tag{F8}$$

Therefore, $\{I_{q_{i,1}}, Z_{q_{i,1}}\} \oplus \{H_{i,t}\}_{t=0}^{m}$ is a valid Pauli envelope for the atom loss of a_i. □

Without loss-resolving readouts, the effect of atom loss is similar to hook errors or single data-qubit errors. Thus, if a code can tolerate $\lfloor (d_h - 1)/2 \rfloor$ hook errors or single data-qubit errors, then it can also tolerate $\lfloor (d_h - 1)/2 \rfloor$ atom loss errors.

Theorem 7 (Tolerance to atom loss without loss-resolving readouts)**.** *Even without loss-resolving readouts, the Mid-SWAP syndrome extraction circuit can correct $\lfloor (d_h - 1)/2 \rfloor$ atom loss errors.*

With loss-resolving readouts, the decoder obtains more information about the atom loss. The decoder (Algorithm 1) will know the index i of the hook error $H_{i,t}$ through the loss-resolving readouts but will be uncertain about the exact time step t. If the decoder guesses the time step as t^*, then the residual error is $H_{i,t} \oplus H_{i,t^*}$.

We define a segment-hook error as an error on a contiguous segment of data qubits in the index order $q_{i,1}, q_{i,2}, \ldots, q_{i,m}$ associated with stabilizer S_i^X. Formally, it is an error of the form

$$H_{i,j_1,j_2} = \prod_{j=j_1}^{j_2} X_{q_{i,j}}. \tag{F9}$$

where $1 \leq j_1 \leq j_2 \leq m$, and contiguity means consecutive indices $j_1, j_1 + 1, \ldots, j_2$ in this order. Let d_s be the minimum number of segment-hook errors whose product implements a nontrivial logical operator.

Theorem 8 (Tolerance to atom loss with loss-resolving readouts)**.** *A code with segment-hook distance d_s can correct any $n_l < d_s$ atom loss errors with loss-resolving readouts.*

Rotated surface codes (as proved in this paper) satisfy the stronger condition $d_s = d_h = d$, thus achieving optimal tolerance to atom loss with the Mid-SWAP technique. The larger family of HGP codes, including unrotated surface codes, which are known to support efficient error correction [48–50], state preparation [51], and logical gates [52–64], have also been shown to achieve the condition $d_s = d_h = d$ [46, 47]. This result is consistent with previous work on toric codes [15]. Thus, they are strong candidates for quantum error-correcting codes for neutral-atom architectures.

Appendix G: Extension to Correlated Atom Loss

We now discuss the tolerance to correlated atom loss. A correlated atom loss is an error in which a physical CZ gate causes both atoms to be lost simultaneously.

Intuitively, correlated atom loss is more destructive than independent atom loss as it introduces high-weight errors easily. However, as we will see, correlated atom loss has the same effective distance as Pauli errors even without loss-resolving readouts.

Lemma 10 (Correlated loss-induced hook error)**.** *Let $\mathcal{C}$ be the Mid-SWAP syndrome extraction circuit from Example 3.*

Let a_i be an X-type ancilla associated with qubits $q_{i,1}, q_{i,2}, \ldots, q_{i,m}$. If a correlated atom loss of a_i with some $q_{i,\tau}$ occurs, then the Pauli envelope of this particular loss configuration has the following form when propagated to the end of this syndrome extraction cycle:

$$E = \{I_{q_{i,\tau}}, Z_{q_{i,\tau}}\} \oplus \{H_{i,\tau}\} \oplus \{\textit{Measurement errors}\}. \tag{G1}$$

where $H_{i,\tau}$ is the hook error

$$H_{i,\tau} = \prod_{j=1}^{\tau} X_{q_{i,j}}. \tag{G2}$$

Proof. The proof follows from the same argument as Lemma 9. □

Theorem 9 (Tolerance to correlated atom loss without loss-resolving readouts)**.** *Even without loss-resolving readouts, the circuit can tolerate $\lfloor (d_h + 1)/2 \rfloor$ correlated atom loss errors.*

With loss-resolving readouts, correlated atom loss provides more information about the locations of the lost atoms. For example, if only one pair of qubits is lost, in the Mid-SWAP circuit, one pair of qubits interacts only once, so we can perfectly identify the locations of the lost atoms. If perfectly identifying the location of the lost atoms were possible, the Pauli envelope contains only a single hook error, and the decoder can correctly decode n_h correlated atom loss errors, meaning $2n_h$ atoms are lost simultaneously.

Although perfectly identifying the location of the lost atoms is not always possible, we can still modify the Envelope-MLE decoder to handle correlated atom loss.

The high-level idea is to jointly minimize the number of correlated losses and independent loss events.

Algorithm 3 (Envelope-MLE decoder with correlated atom loss)**.** *Before receiving the detector patterns, the decoder first performs the following steps as preprocessing:*

1. ***Find the Pauli envelope for each loss-resolving readout:*** *For each measured atom m, find the Pauli envelope E_m of the single-loss readout $r^{(m)}$ as in Algorithm 1.*

2. ***Find the Pauli envelope for each correlated loss-resolving readout:*** *For each pair of atoms (m,n), $m<n$, that can be lost together, find the Pauli envelope $E_{m,n}$ of the readout $r^{(m)} \oplus r^{(n)}$ corresponding to a single correlated atom loss, according to Lemma 2.*

3. ***Assign variables for loss patterns:*** *For each pair $(\mathcal{D}_{m,j}, \mathcal{O}_{m,j}) \in \mathcal{S}(E_m, \emptyset)$, assign a binary variable $x_{m,j} \in \{0,1\}$, indicating whether the loss pattern is selected ($x_{m,j}=1$) or not ($x_{m,j}=0$).*

4. ***Assign variables for correlated loss patterns:*** *For each pair $(\mathcal{D}_{(m,n),j}, \mathcal{O}_{(m,n),j}) \in \mathcal{S}(E_{m,n}, \emptyset)$, assign a binary variable $x_{(m,n),j} \in \{0,1\}$, indicating whether the correlated loss pattern is selected ($x_{(m,n),j}=1$) or not ($x_{(m,n),j}=0$).*

5. ***Assign variables for Pauli errors:*** *For each Pauli error mechanism i in the circuit, assign a binary variable $y_i \in \{0,1\}$, indicating whether the Pauli error mechanism is selected ($y_i=1$) or not ($y_i=0$).*

After receiving the detector pattern and loss-resolving readout, the decoder performs the following steps:

1. ***Enforce constraints:***

   (i) Detector satisfaction: *For each detector u, enforce*

$$\sum_i D_{i,u} y_i + \sum_{m,j} D_{m,j,u} x_{m,j} + \sum_{(m,n),j} D_{(m,n),j,u} x_{(m,n),j} \equiv s_u \pmod 2, \tag{G3}$$

   where $D_{i,u} \in \{0,1\}$ indicates whether Pauli error i flips detector u, $D_{m,j,u} \in \{0,1\}$ indicates whether loss pattern (m,j) flips detector u, and $s_u \in \{0,1\}$ is the observed value of detector u.

   (ii) Loss exclusivity: *For each readout bit r_m, where $r_m=1$ indicates that atom m is reported as lost, enforce*

$$\sum_j x_{m,j} + \sum_{\substack{(a,b),j \\ m\in\{a,b\}}} x_{(a,b),j} = r_m, \tag{G4}$$

   ensuring exactly one detector pattern is selected per loss event.

2. ***Minimize loss error count and then Pauli error count:*** *Solve the MILP:*

$$\min_{x,y} \left(\sum_{m,j} x_{m,j} + \sum_{(m,n),j} x_{(m,n),j} + \epsilon \sum_i w_i y_i \right) \tag{G5}$$

   subject to the above constraints, where ϵ is a small positive constant ensuring the loss error count is minimized first.

3. ***Predict logical observable:*** *Let $x^*_{m,j}, y^*_i$ be the optimal solutions to the MILP. Calculate the predicted logical flip:*

$$\mathcal{O}_{pred} = \bigoplus_i y^*_i \mathcal{O}_i \oplus \bigoplus_{m,j} x^*_{m,j} \mathcal{O}_{m,j} \oplus \bigoplus_{(m,n),j} x^*_{(m,n),j} \mathcal{O}_{(m,n),j} \tag{G6}$$

   where $\mathcal{O}_i$ is the observable flip caused by Pauli error i.

For evaluation, we consider only atom loss after each $CNOT$ gate and do not consider Pauli errors. For an error rate p and correlated loss contribution η, each atom has probability $p(1-\eta)/2$ of being lost independently and probability $p\eta/2$ of being lost together with its partner. Thus, the marginal probability of each atom being lost remains $p/2$. As shown in Fig. 18, the threshold

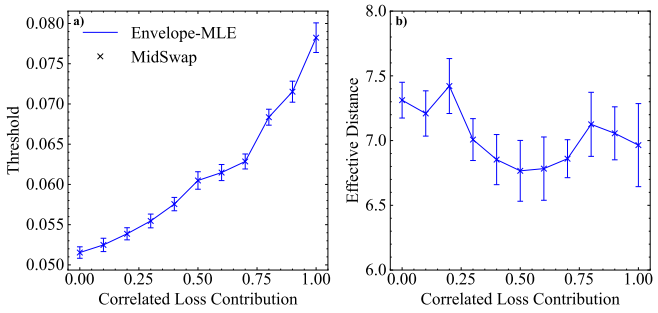


FIG. 18: (a) Threshold versus correlated-loss contribution; (b) effective distance at code distance 7 versus correlated-loss contribution.

increases from 0.05 to 0.08 as the correlated loss contribution η increases from 0 to 1. Furthermore, the effective distance remains approximately 7.

In summary, without loss-resolving readouts, both independent atom loss and correlated atom loss behave similarly to hook errors. With loss-resolving readouts, independent atom loss behaves similarly to a segment-hook error with known locations, while correlated atom loss performs at least as well as independent atom loss and provides a higher threshold by supplying more information about the locations of the lost atoms. This finding corroborates a recent independent study [65], which

also observes that correlated atom loss can be beneficial for error correction.

We leave for future work a careful analysis of the interplay among regular Pauli errors, independent atom loss, and correlated atom loss errors, as well as a threshold theorem and relaxation of the assumptions that each ancilla qubit is only associated with one data qubit under alternating syndrome extraction orders.

Appendix H: Teleportation-based Syndrome Extraction Circuit

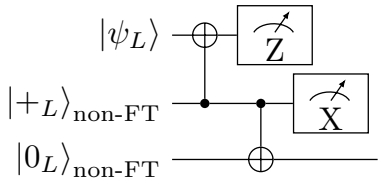


FIG. 19: Teleportation-based syndrome extraction circuit proposed in Ref. [14].

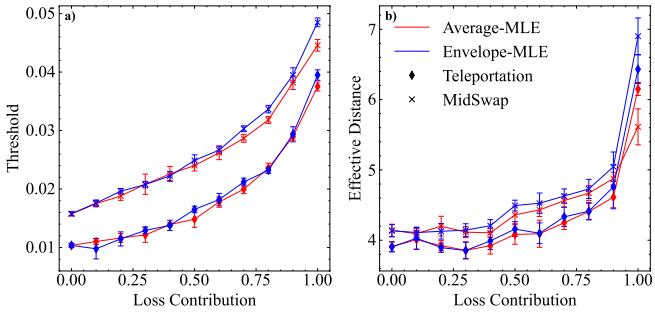


FIG. 20: Performance comparison between the teleportation circuit and the Mid-SWAP syndrome extraction: (a) threshold; (b) effective distance at code distance 7.

Ref. [14] also proposes a teleportation-based syndrome extraction circuit, which can be viewed as a modification of the Steane error correction circuit. The circuit is shown in Fig. 19. The two ancilla logical qubits are prepared non-fault-tolerantly, meaning that only one syndrome extraction round is performed. Although Ref. [14] shows that preparing the ancilla qubits fault-tolerantly can achieve a much higher threshold, this approach becomes a bottleneck when combined with transversal logical circuits. Therefore, we only consider the non-fault-tolerant preparation of ancilla qubits here, which takes constant time and provides full circuit-level distance.

We exclude the teleportation-based syndrome extraction circuit from the main text due to its significantly inferior performance compared to both Mid-SWAP syndrome extraction and SWAP syndrome extraction. The teleportation circuit exhibits two major disadvantages: (i) substantially lower threshold and effective distance, and (ii) considerably higher space-time overhead. Fig. 20 quantifies these performance gaps.

To ensure a fair comparison, we exclude idling errors from the teleportation circuit because its non-fault-tolerant state preparation is significantly affected by idling errors, while retaining the original error model for Mid-SWAP syndrome extraction.

The teleportation circuit achieves a threshold that is 0.5% to 1% lower than that of Mid-SWAP syndrome extraction, along with a slightly lower effective distance.

Appendix I: Measurement and reset error sensitivity

Both SWAP syndrome extraction and Mid-SWAP syndrome extraction exhibit different error-propagation characteristics compared with the standard surface-code syndrome extraction circuit under pure Pauli noise. The standard surface-code circuit confines both measurement and reset errors to time-like edges, providing better tolerance to these errors. In contrast, SWAP syndrome extraction propagates measurement errors to space-like edges, while Mid-SWAP syndrome extraction propagates reset errors to space-like edges.

Fig. 21 examines how logical error rates depend on the relative magnitudes of reset and measurement error rates for both circuits. When the sum of reset and measurement error rates is fixed, the results reveal that the logical error rate of Mid-SWAP syndrome extraction increases as the reset error rate increases, whereas SWAP syndrome extraction's logical error rate increases as the measurement error rate increases.

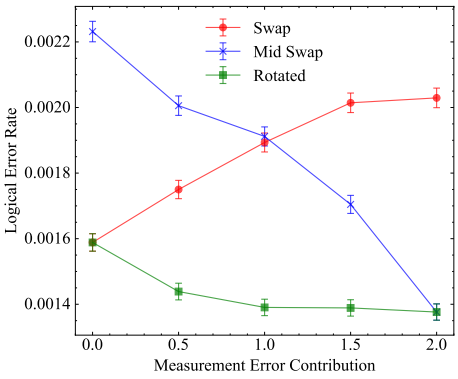


FIG. 21: Logical error rate behavior of Mid-SWAP syndrome extraction, SWAP syndrome extraction, and standard surface code syndrome extraction with varying reset and measurement error rates. Simulations are performed at distance $d = 5$ with error rate $p_{\text{pauli}} = p_{\text{loss}} = 0.008$. We scale the reset error rate to $0.008x$ and measurement error rate to $0.008(2 - x)$, where x ranges from 0 to 2. The Mid-SWAP syndrome extraction is more sensitive to reset errors, whereas the SWAP syndrome extraction is more sensitive to measurement errors.

Appendix J: Search for optimal hyperparameters

For Envelope-Matching decoder, our theoretical analysis derives reweighting coefficients of 0.5 for space-like edges and 0.25 for time-like edges.

With finite distances and error rates, these values might not give the best practical performance. Furthermore, any value in the range of $(0, 0.25)$ for time-like edges gives the same asymptotic performance.

Therefore, we view the reweighting coefficients as hyperparameters and empirically optimize them for the Envelope-Matching decoder in SWAP syndrome extraction and Mid-SWAP syndrome extraction and show the result at code distance $d = 13$ as a function of loss contribution in Fig. 22.

The results show that the empirically optimized hyperparameters closely match the theoretical predictions for Mid-SWAP syndrome extraction, confirming the correctness of our analytical approach.

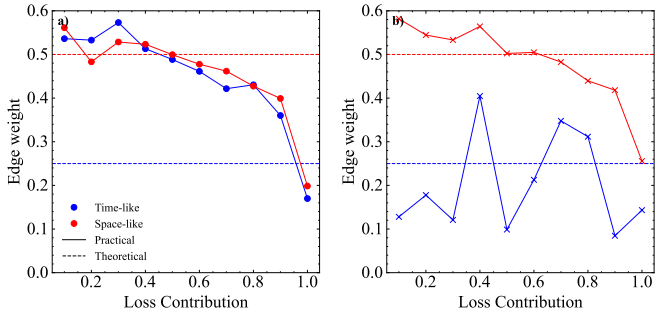


FIG. 22: Optimal hyperparameters for the Envelope-Matching decoder in (a) SWAP syndrome extraction and (b) Mid-SWAP syndrome extraction at code distance $d = 13$ as a function of the loss contribution. For Mid-SWAP syndrome extraction, the empirically optimized values closely match the theoretically predicted values of 0.5 for space-like edges and 0.25 for time-like edges, validating our analytical approach across different loss rates.

Appendix K: Nonlinearity of atom loss events

Our Envelope-MLE decoder (Algorithm 1) converts atom loss events to Pauli envelope errors rather than directly enumerating loss-induced detector-observable pairs as in Ref. [14]. This conversion is not only essential for our proof but also for the practical performance of the decoder.

Due to the nonlinearity of atom loss, when multiple losses occur simultaneously but are not modeled properly, the Envelope-MLE decoder has to use multiple Pauli errors to model the loss-induced errors, significantly degrading the performance.

Fig. 23 empirically validates this argument by comparing the Envelope-MLE decoder with the correct implementation and a modified version where the first step is removed, and the detector-observable pairs are enumerated by removing the Clifford gates affected by the loss. For SWAP syndrome extraction, the naive gate-removal approach yields significantly higher logical error rates, confirming that loss nonlinearity cannot be ignored in this circuit. However, for Mid-SWAP syndrome extraction, both approaches yield nearly identical performance, indicating that atom losses behave almost linearly in this architecture. We conjecture that this near-linearity also contributes to the superior performance of Mid-SWAP syndrome extraction over SWAP syndrome extraction when using Envelope-MLE decoder.

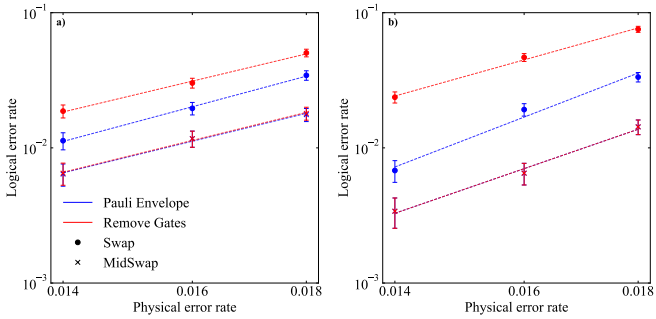


FIG. 23: Impact of the Pauli envelope on decoding performance for SWAP syndrome extraction and Mid-SWAP syndrome extraction at code distances (a) 5 and (b) 7. To emphasize the effect, we set the loss contribution to 1 in all cases.

Appendix L: Attribution of the gain and the fitted exponents of the baselines

Circuit versus decoder. Fig. 10 contains all four combinations of decoder (Average-MLE decoder, Envelope-MLE decoder) and circuit (SWAP syndrome extraction, Mid-SWAP syndrome extraction) under the same noise model, distances, and sampling. Table III collects the numbers that separate the two contributions, and Fig. 1 shows the logical error rates of the four combinations at one pure-loss operating point.

TABLE III: Threshold and effective distance of the four decoder–circuit combinations. Thresholds are fitted to $d \in \{5, 7, 9\}$; effective distances are at $\eta = 1$. Parentheses give the uncertainty in the last digits.

Decoder	Circuit	$p_{\text{th}}^{\eta=1}$ (%)	$p_{\text{th}}^{\eta=0.5}$ (%)	$d_{\text{eff}}^{d=9}$	$d_{\text{eff}}^{d=11}$
Average-MLE	SWAP	3.35(3)	2.11(3)	6.48(9)	8.67(14)
Average-MLE	Mid-SWAP	4.18(9)	2.27(5)	6.77(10)	8.57(25)
Envelope-MLE	SWAP	3.89(8)	2.13(11)	7.57(14)	10.40(22)
Envelope-MLE	Mid-SWAP	4.75(5)	2.36(6)	8.83(18)	11.31(12)

Stabilizer order. The circuits alternate the CNOT order between rounds so that every boundary data qubit is

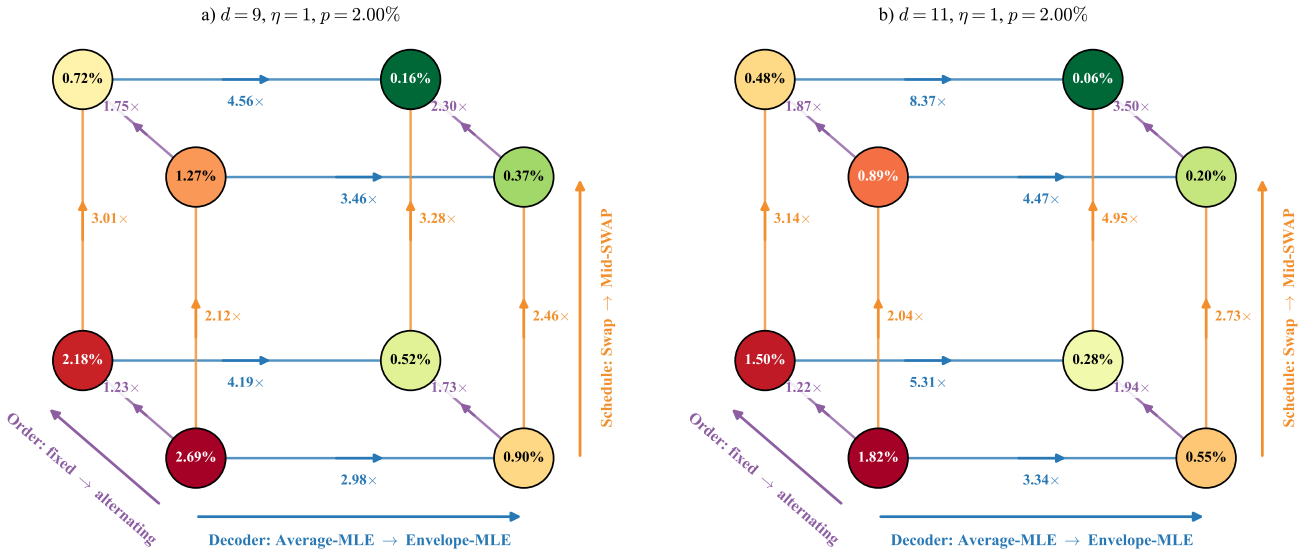


FIG. 24: Logical error rate of eight configurations at pure loss ($\eta = 1$) and $p = 2.0\%$ for (a) $d = 9$ and (b) $d = 11$, along three axes: decoder (Average-MLE decoder or Envelope-MLE decoder), schedule (SWAP syndrome extraction or Mid-SWAP syndrome extraction), and stabilizer order. The front face uses the alternating order of this paper (the CNOT order reversed in every other round) and repeats Fig. 1; the back face uses a fixed order made possible by one extra replenishment ancilla per data qubit on the bottom and right boundaries. Arrows point from the baseline to the improved setting and are labeled with the matched error-reduction factor.

replenished (Section VII); the alternative is a fixed order with one extra replenishment ancilla for each boundary data qubit that the fixed order misses, $O(d)$ in total. Fig. 24 compares the two remedies as the third axis of a configuration cube whose front face is Fig. 1: the alternating configuration has the lower logical error rate for every decoder and circuit, by factors of 1.2 to 2.3 at $d = 9$ and 1.2 to 3.5 at $d = 11$. Since this comparison changes both the order and the boundary construction, we also ran a control in which only the order changes, at the fixed geometry of the back face and with the Average-MLE decoder. Alternating the order lowers the logical error rate by a factor that grows with the distance: at pure loss and $p = 2.38\%$ it is 1.02, 1.30, 1.37, and 1.50 for $d = 3, 5, 7, 9$; at $\eta = 0.5$ it grows from 1.0 at $d \leq 5$ to 1.30 at $d = 9$.

This effect is not captured by the average atom lifetime, the figure of merit used in Ref. [14] to compare loss-detection circuits, where a shorter lifetime is expected to give a lower logical error rate. In units of CNOT layers, the average lifetime is $8 + o(1)$ under the alternating order and $8 - o(1)$ under the fixed order with boundary ancillas, the $o(1)$ being boundary corrections, so this metric slightly favors the fixed order, opposite to the observed ordering. The alternation is therefore a genuine, distance-dependent advantage on the planar code, consistent with the distance-preserving role of alternating orders in Refs. [66, 67]; a proof is left to future work.

Fitted exponents of the baselines. At the error rates and distances studied, the fitted d_{eff} of the Average-MLE decoder (Table III) and of the Marginal-Matching decoder (Fig. 12) lie above their proven $d_{\text{loss}} \sim d/2$, and that of the Envelope-Matching decoder above its proven $2d/3$. This is expected and does not contradict our theoretical bounds, which hold for sufficiently small error rates. At finite error rates and code sizes, the logical error rate can fall faster than the asymptotic distance scaling, a behavior known as the waterfall effect [68]. Several finite-size effects contribute to the steeper fitted slopes. (i) Failures heavier than the minimum can contribute at the finite error rates used in the fit, increasing the slope above its low-error limit. (ii) $d_{\text{loss}} \sim d/2$ means $d/2 \pm O(1)$, and at $d \leq 13$ an offset of one or two is a sizable fraction of $d/2$. (iii) The minimum-weight failures of Appendix E 2 require each loss at a specific circuit location and, for the Average-MLE decoder, that it triggers no detector, so each event carries a constant factor $c < 1$. Their total probability can therefore be small relative to that of more numerous heavier failures. At these sizes, the practical separation between the baselines and the Envelope-Matching decoder is smaller than the proven bounds suggest; its practical gains are the higher threshold and the stability as $\eta \to 1$ (Section VII).

Appendix M: Numerical details

1. Sampling procedure

All sampling is done with Stim [37]. Stim does not support atom loss natively, so we extend it in the following way.

We first scan through the circuit, and collect the locations where an atom can be lost, and the gates that each atom loss location will affect. For each sample, we draw a loss configuration by triggering each atom loss independently with the given error rate, and remove the affected gates.

After that, we get a Stim circuit with some gates removed. We then insert Pauli errors into the circuit. An atom whose partner is lost undergoes an idling error. One technical point remains: after the gates are removed, the observable is no longer deterministic even without Pauli errors, and Stim rejects such a circuit by default. We therefore convert the observable into a detector and sample with Stim's `allow_gauge_detectors` option. After that, we sample the raw measurement outcomes, mask out those of the lost atoms so that they leak no information to the decoder, and convert them into detectors.

Since each new loss configuration requires its own compiled circuit, which can become a bottleneck when using matching-based decoders, we use a two-stage sampling procedure. For each loss configuration, we sample two Pauli-noise shots from the circuit, which roughly speeds up the sampling by a factor of two but introduces a small correlation between the two shots.

2. Details on solver timeouts

Every shot decoded by the Average-MLE decoder or the Envelope-MLE decoder is a separate Gurobi MILP, solved with a 60-second time limit and a relative MIP gap of 1%. If the time limit is reached, the best feasible solution found so far is used as the correction; in extremely rare cases, the solver is unable to find any feasible solution, in which case the shot returns a zero correction by default.

In our experiments, time-limit hits are rare. Across all $d = 9$ runs they affect fewer than 0.02% of the shots, except for the near-threshold data of the Envelope-MLE decoder on the Mid-SWAP syndrome extraction, where 1.2% of the shots reach the time limit.

To bound the effect of these time-limit hits, we re-decoded 200,000 near-threshold $d = 9$ shots of the Envelope-MLE decoder on the Mid-SWAP syndrome extraction ($\eta \in \{1, 0.9, 0.8, 0.7\}$, five values of p each, 10,000 shots per point) from identical cached samples under the two solver settings of Table IV. Raising the time limit to 300 s removes all time-limit hits but changes the correction of only 51 shots and the number of logical failures by five, which is not statistically significant.

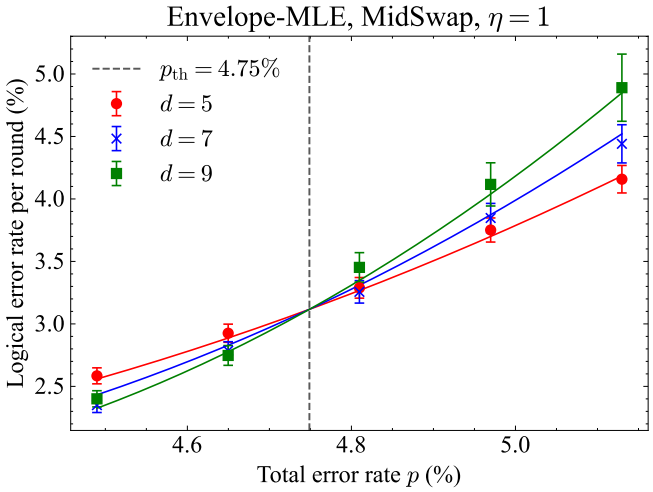


FIG. 25: Threshold crossing of the Envelope-MLE decoder on Mid-SWAP syndrome extraction at loss contribution $\eta = 1$ for $d \in \{5, 7, 9\}$: logical error rate per round versus total error rate p. The solid curves are the common finite-size-scaling fit $P_{\rm LER} = a + bx + cx^2$ with $x = (p - p_{\rm th})\, d^{1/\nu}$, and the dashed line marks the fitted threshold. Error bars are one standard error; the $d = 9$ points use 20,000 shots each.

TABLE IV: Re-decoding the same 200,000 near-threshold $d = 9$ shots of the Envelope-MLE decoder on the Mid-SWAP syndrome extraction under two Gurobi settings. The "Changed corrections" row counts shots whose correction differs from the 60 s, 1% MIP gap setting used in the paper.

Time limit	60 s	300 s
Relative MIP gap	1%	1%
Time-limit hits	126	0
Logical failures	89,297	89,302
Changed corrections	–	51

3. Threshold crossings

We show an example of the fitted thresholds to show the threshold crossing behavior. Fig. 25 shows the underlying crossing for the Envelope-MLE decoder on the Mid-SWAP syndrome extraction at $\eta = 1$.

Appendix N: Summary of Notation

Table V collects the main symbols, grouped by topic. Auxiliary symbols used only in individual proofs are defined locally.

Symbol	Meaning	Introduced in
Circuits and error configurations		
$\mathcal{C}$, $\mathcal{C}_i$	Clifford circuit; auxiliary circuits used in proofs	Section III, Appendix A 2
p	Pauli error configuration: a Pauli operator assigned to each faulty space-time location	Section III
l, l_i	loss configuration: set of space-time locations at which atoms are lost; single-atom loss configuration	Section III
$\emptyset$	the loss-free configuration	Definition 2
$\oplus$	Pauli product with phases discarded; XOR for bits or chains; union for loss configurations or readouts	Section III
P, L	sets of Pauli error configurations and of loss configurations	Section III
$\mathcal{P}_p$, $\mathcal{P}_l$	distributions of Pauli error and loss configurations	Section III
R, $\mathcal{N}$	reset gate; completely depolarizing channel modeling the random bit of a lost atom	Appendix A 2
Readouts		
$\tilde{b}_m \in \{0, 1, \perp\}$	physical three-outcome readout of atom m; $\perp$ means the atom is lost	Section III
b_m	binary bit used to evaluate detectors and observables; uniformly random if $\tilde{b}_m = \perp$	Section III
$r = (r_m)_m$	loss-resolving readout, with $r_m = 1$ if and only if $\tilde{b}_m = \perp$	Section III
$\mathcal{R}$	readout map, $r = \mathcal{R}(l)$	Section III
$\mathcal{L}(r)$	preimage of r: all loss configurations with $\mathcal{R}(l) = r$	Section III
$r^{(m)}$	readout in which only atom m is reported as lost	Lemma 3
Detectors, outcomes, and Pauli envelopes		
$\mathcal{D}$, $\mathcal{O}$	detector pattern; logical observable flip	Section II, Section III
$\mathcal{S}(p, l)$, $\mathcal{S}(P, L)$	support of detector-observable outcomes; union over sets of configurations	Section III
E, $E(l)$, $E(r)$	Pauli envelope (a set of Pauli error configurations) of a loss configuration or readout	Definition 2
Π_i, Π_q^τ	Pauli set $\{I, X, Y, Z\}$ at space-time location i (on qubit q at time τ)	Lemma 2, Appendix F
Dec	decoder mapping $(\mathcal{D}, r)$ to a predicted observable	Lemma 1
P_{LER}, P_{fail}	logical error rate; failure probability with Pauli envelope	Definition 1, Section III
Distances, error rates, and counts		
d, d_{loss}	code distance; loss distance	Definition 1
$\sim f(d)$	$f(d) \pm O(1)$	Section I
d_{eff}	fitted effective distance (exponent of the low-error scaling)	Section VII
p_{loss}, p_{pauli}, p_{readout}	physical loss, Pauli, and loss-readout error rates	Definition 1, Section VII
p, η (Section VII)	total physical error rate and loss contribution: $p_{\mathrm{loss}} = \eta p$, $p_{\mathrm{pauli}} = (1 - \eta)p$	Section VII
p, η (Appendix G)	loss-rate parameter (marginal loss probability $p/2$ per atom) and correlated-loss contribution	Appendix G
$\bar{p}$	unified rate in scaling proofs, e.g. $\max(p_{\mathrm{loss}}, \sqrt{p_{\mathrm{pauli}}})$	Appendix C 4
p_{th}	threshold error rate	Definition 1
n_l, n_p, n_f	numbers of atom losses, Pauli errors, and undetected (false-negative) losses	Theorem 2, Theorem 5
Λ	error suppression factor	Section VII
Decoders		
E_m, $E_{m,n}$	Pauli envelope of $r^{(m)}$; of a correlated loss of atoms m and n	Algorithm 1, Algorithm 3
$(\mathcal{D}_{m,j}, \mathcal{O}_{m,j})$	j-th detector-observable pair in $\mathcal{S}(E_m, \emptyset)$	Algorithm 1
$x_{m,j}$, $x_{(m,n),j}$	binary envelope-selection variables	Algorithm 1, Algorithm 3
y_i, w_i, w	binary variable and log-likelihood weight of Pauli error mechanism i; common weight	Algorithm 1
$\lambda_{m,j}$	negative log-probability of envelope pattern j for atom m reported as lost in the decoder model	Algorithm 1
u, s_u	detector index; observed value of detector u	Algorithm 1
$D_{i,u}$, $D_{m,j,u}$	whether Pauli error i / loss pattern (m, j) flips detector u	Algorithm 1
$G = (V, \mathcal{E})$, w_e	matching graph and its edge weights	Algorithm 2
$S_1, \ldots, S_4$	detector pairs (edges) that an ancilla loss can trigger in Mid-SWAP syndrome extraction	Lemma 4
Proofs		
$F_{m,j}$, $\partial F_{m,j}$	representative chain for $(\mathcal{D}_{m,j}, \mathcal{O}_{m,j})$; its detector boundary $\mathcal{D}_{m,j}$	Appendix C 2
$C = C_P \oplus C_L$	difference of actual and decoded representative chains, split into Pauli and loss components	Appendix C 2
F, $\widehat{F}$	actual error chain and MWPM correction chain in the matching graph	Appendix D 1

TABLE V: Summary of notation.